Original Paper

Brain, Behavior and Evolution

Brain Behav Evol 1998;51:59–89

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Comparative Morphometry of Mammalian Central Auditory Systems: Variation in Nuclei and Form of the Ascending System

Key Words
Hearing
Allometry
Marsupials
Placentals
Evolution

Abstract
The volumes of the ten largest subcortical auditory nuclei were measured individually in a sample of 53 mammals, including 16 Australian and four American marsupials. The nuclear sizes relative to the total of subcortical auditory tissue were normalized and then analyzed individually for statistically reliable deviations. The overall form of the entire system of ten nuclei and two nuclear subsystems (cochlear nuclei, superior olives) were also analyzed for similarities and notable deviations among the animals. The results show that the absolute size of the auditory system varies more than 139-fold among the 53 mammals (with moles the smallest and humans the largest). Log auditory system volume and log brain weight are closely correlated ($r = 0.903$, $p < 0.0001$). Bats, kangaroo rats, marmosa opossums, and Norway rats have the largest auditory systems relative to their brain size, while humans have the smallest by far. The other primates also have auditory system/brain size ratios smaller than the sample average, suggesting that the condition in humans is one result of an expansion of non-auditory brain parts rather than a reduction of the auditory system over geological time. The relative sizes of the ten nuclei are well ordered, with the inferior colliculus the largest nucleus by far and medial superior olive the smallest. Because the size of the superior olives, collectively, is reliably related to the size of anteroventral cochlear nucleus ($r = 0.744$, $p < 0.001$), and not to the size of dorsal cochlear nucleus, the interconnectivity of the subcortical auditory system is probably a factor in the size of the nuclei. In its overall form, the subcortical auditory system is highly similar among mammals, with an average correlation across nuclei of 0.923. This high value means that the overall form of the system has been relatively stable over geological time. The animals with least deviation from the average form are ring-tailed possums, bandicoots, and yellow-bellied gliders, all marsupials. Those with the most unusual forms are mice, bats, and kangaroo rats, all placentals.

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Abbreviations

AVCN	Anteroventral Cochlear Nucleus
CN	Cochlear Nucleus (DCN + AVCN + PVCN)
DCN	Dorsal Cochlear Nucleus
DLL	Dorsal Nucleus of the Lateral Lemniscus
IC	Inferior Colliculus
LSO	Lateral Superior Olivary Nucleus
MG	Medial Geniculate
MSO	Medial Superior Olivary Nucleus
MTB	Medial Nucleus of the Traezoid Body
PVCN	Posteroventral Cochlear Nucleus
SO	Superior Olives (LSO + MSO + MTB)
VLL	Ventral Nucleus of the Lateral Lemniscus

Introduction

There are at least 22 auditory nuclei known to be present between the ear and auditory cortex on each side of the brain (i.e., 44 in total from cochlear nuclei through medial geniculate). Eight of the 22 nuclei on each side surround the superior olives – so-called periolivary nuclei. They have been shown to originate olivocochlear and olivocochlear nucleus fibers, and five of these eight probably serve largely this 'descending' role [Warr, 1975, 1978, 1982; Irvine, 1986; Spangler et al., 1987]. Subtracting this number from the total leaves all or part of 17 nuclei to make up the 'ascending' auditory system.

After a century of physiological and psychoacoustical research into the ascending auditory system, only three of the 17 nuclei – the so-called superior olives – and two others by hodological inference have begun to be understood in terms of auditory perception or hearing. We now know that one major contribution of the superior olives, i.e. the lateral and medial superior olivary nuclei and the medial nucleus of the trapezoid body (LSO, MSO, and MTB), is the analysis of interaural time and intensity differences for sound-source localization [Boudreau and Tsuchitani, 1968, 1970; Masterton et al., 1975; Masterton, 1992]. Because these three nuclei receive their afferents mostly from the anteroventral cochlear nucleus (AVCN) and project most of their efferents to the inferior colliculus (IC), these two nuclei are also implicated in the same process. It should be noted that this nuclear mechanism for sound-source localization accounts for, at most, 5 of the 17 nuclei in the ascending system [Masterton, 1992].

Reasonable hypotheses regarding the contributions of many of the remaining nuclei can be deduced by the nature of their afferent supplies. However, in the past, this type of reasoning has rarely been fruitful for predicting physiological characteristics beyond the most general – for example, 'tonotopicity' or 'binaural sensitivity'. Hence, these hypotheses fall short of generating more global hypotheses sufficiently explicit to be useful for psychoacoustical experimentation. In general, specific roles of the majority of auditory nuclei and their intrinsic networks have yet to be proposed.

In the hope of providing some fresh clues that might be used to generate plausible hypotheses for further exploration of physiological, psychoacoustical, or behavioral contributions, we have reverted to the method of comparative morphometry focused on the nuclei in the ascending system [Thompson, 1917; see also Armstrong, 1985; Fleagle, 1985; Rohlf and Marcus, 1993; Johnson and Kirsch, 1993]. Although it can be argued that a more appropriate unit of analysis is likely to be found at a more refined level (e.g., clusters of similar cell types, numbers of neurons with similar afferents, or total cellular membrane available for synapses, etc.), we have chosen arbitrarily to begin at the level of nuclei in their entirety mostly because more refined levels are not yet practicably measurable over suitably large samples of animals. In addition, whole nuclei are almost certainly homologous (in the sense that they were present in the common ancestry of the mammals in the sample), whereas their more refined subdivisions or constituent cell-types may not be homologous in the same sense.

Therefore, the proposition on which the present report is based is that the deviation from the mean in relative size of a central nervous system nucleus is an indication of its relative usefulness to its host. If this proposition is tentatively accepted, it follows that the form of a multi-nuclear neurological system – the relative sizes of its several constituent nuclei – might be one of the more sensitive indicators of the presence of special selective pressures in an animal's environment and vice versa. Therefore, the identification of unusual deviations in the size of the constituent nuclei or in the overall form of the auditory system, especially if they prove to be paralleled by unusual acoustical demands, might provide clues to the functional contributions of the nuclei. In this first report, we identify some animals that have either individual auditory nuclei or overall forms of the ascending auditory system, cochlear nuclei, or superior olives sufficiently unusual as to be properly called 'deviant'.

The first step in the comparative method used here is to establish the usual form of the system and its range of variation. Although this 'usual' or average form can be used for taxonomic, phylogenetic, and ecological analyses, we use it here only for a practical purpose: it provides a means by which a deviant nucleus, a deviant cluster of nuclei, or a

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deviant form of the entire system, can be identified. Accordingly, we have measured the volume of the 10 largest nuclei – the 10 nuclei which are large enough to be reliably measured – in the ascending system of 53 species of mammals. The relative sizes of each of the 10 nuclei, viewed individually, are used to identify the animals with one or another deviant nucleus. The sizes of the same nuclei, viewed collectively, are then used as a profile of the overall 'form' of each animal's auditory system. The collection of these forms, in turn, provides the statistical basis for describing the usual form of the system, the range of its variation, and some of its notable deviations.

Materials and Methods

The brain sections used for volumetric measurements were all Nissl-stained materials. They were obtained from five different sources: University of Wisconsin (courtesy of W. Welker); Monash University, Australia (courtesy of L. Aitkin and J. Nelson); Armed Forces Institute of Pathology (courtesy of E. Armstrong); University of Toledo (courtesy of R.S. Heffner); and our own materials at Florida State University (see table 1).

Although well-established and routine histological procedures were followed for preparing each set of brain sections, three different embedding and sectioning techniques were represented. Celloidin embedding was used in 31 of the brains, paraffin embedding in 6, and frozen sectioning with gelatin embedding in the remaining 16. The effects of these differing procedures on the raw measurements of nuclear volume and the steps taken to minimize the inherent error are presented in the results.

Identification of Auditory Nuclei

In order to quantify the form of the ascending auditory system, the ten largest nuclei in the ascending auditory system were chosen for volumetric measurement: dorsal, posteroventral and anteroventral cochlear nuclei (DCN, PVCN, AVCN), lateral and medial superior olivary nuclei (LSO, MSO), medial nucleus of the trapezoid body (MTB), ventral and dorsal nuclei of the lateral lemniscus (VLL, DLL), inferior colliculus (IC), and medial geniculate (MG) (see fig. 1). The relative sizes of these ten nuclei are used first to identify deviant nuclei and, again, to profile the 'form' of the entire system in each animal.

Calculation of Nuclear Volumes

Equidistant transverse sections of brain tissue were systematically projected onto paper, and after the appropriate magnification was determined, the auditory nuclei were outlined. After each of the nuclei were outlined on both sides of the brain in each section, the area of the individual nuclei in each drawing was determined with a computerized planimeter ('Image 1.45', NIH). For each nucleus on each side, the individual measurements of area were summed and multiplied by the distance between sections [$V = d(\Sigma A)$; where d = intersection distance; and ΣA = the sum of the nuclear areas in each section]. This three-dimensional value was then corrected for the drawing's magnification and to adjust the measurement units from μm^3 to mm^3 throughout. The left and right volumes for each nucleus were then averaged. This final value (in mm^3) was used as the raw estimate of the in vitro volume of each nucleus.

Statistical Analysis

After raw volume measurements were obtained for each of the 10 nuclei in each of the 53 animals, the sum of the 10 nuclei in each animal was obtained. This sum for each animal, referred to as 'system total' or 'total' in the tables and graphs to follow, was used as the divisor of each of the 10 nuclei in order to obtain the *relative size* or percentage of each nucleus within each animal. Artithmetical calculations and statistical descriptions and inferences were then performed on these data after appropriate transformation (log V) and normalization [$V' = (V - \mu)/\sigma$].

Species Represented

In all, the measurements were obtained for 53 individual species representing 53 genera, 34 families, 13 orders, and 2 subclasses of Mammalia. The choice of the animals in the sample was *not* random for reasons discussed below. Instead, the selections were based mostly on taxonomic and paleontological grounds. Therefore, there is a large overweighting of marsupials and a small overweighting of primates. There is also a great underweighting of Chiroptera.

Results and Discussion

The technique for estimating the volumes of the auditory nuclei introduces several unusual kinds of measurement error. For this reason, we began with four preliminary inquiries to estimate the total error in the data. Because the outcome of these inquiries helps to explain the restrictions placed on statistical analysis and the constraints on the conclusions, a brief description is useful.

Measurement Error

There are four unusual sources of measurement error in the technique used here: (1) the validity error introduced by using in vitro rather than in vivo material; (2) the validity error in using only one animal to represent a species or genus; (3) the validity and reliability errors in the judgement and mechanics of outlining the nuclei in each tissue section; and (4) the reliability error in approximating a nuclear volume from a systematic but incomplete sample of cross-sectional areas.

The first two of these sources of error are real, pervasive, and unavoidable. They are discussed first, along with the strategy adopted to minimize their effect on the conclusions. The other two sources of error proved to be measurable with the morphometric data themselves. They are described later.

In vitro versus in vivo Volumes

The brain sections used for this study were all Nissl-stained materials. Most of the brains (31/53) had been per-

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Table 1. In vitro volumes (in mm^3) of 10 largest nuclei (between vertical lines) in ascending auditory system of 53 mammals, with body and brain weights (in gms.) and total of the 10 nuclear volumes ('total')

Animal No.	Common name	Genus and species	Source	Body gm	Brain gm	AVCN	PVCN	DCN	MTB	LSO	MSO	VLL	DLL	IC	MG	Total
1	Armadillo	*Dasypus novemcinctus*	FSU	5,000	7.54	3.70	2.45	3.82	0.575	0.550	0.250	1.11	0.980	27.6	8.15	49.1
2	Bandicoot	*Isoodon macrourus*	Monash	4,695	6.16	0.500	0.940	1.19	0.300	0.240	0.100	0.975	0.155	9.40	2.12	15.9
3	Bat, little brown	*Myotis lucifugus*	FSU	8.61	0.18	0.552	0.089	0.135	0.112	0.125	0.064	0.095	0.039	0.630	1.14	2.98
4	Beaver	*Castor canadensis*	U. Wisc.	17,600	36.5	2.60	0.965	3.61	1.22	0.378	0.488	3.13	0.572	36.7	19.2	68.9
5	Bushbaby	*Galago senegalensis*	FSU	210	3.2	0.384	0.430	0.397	0.051	0.098	0.083	0.179	0.076	1.21	0.656	3.56
6	Capybara	*Hydrochaeris hydrochaeris*	U. Wisc.	7,800	51	4.91	1.30	6.78	2.89	1.22	0.903	4.22	1.80	48.9	41.9	115
7	Cat	*Felis catus*	FSU	3,068	19.08	3.23	6.71	5.50	1.26	1.56	1.76	3.77	1.33	38.0	40.9	104
8	Colocolo	*Dromiciops australis*	Monash	26	0.69	0.079	0.165	0.279	0.059	0.038	0.029	0.107	0.018	1.50	0.380	2.65
9	Chinchilla	*Chinchilla laniger*	FSU	(580)		0.63	1.83	3.06	0.28	0.33	0.22	1.10	0.16	7.17	4.73	24.11
10	Deer	*Odocoileus virginianus*	U. Wisc.	40,000	160	9.78	4.06	11.6	3.61	0.621	1.16	2.53	1.83	78.5	61.3	175
11	Dunnart, darling downs	*Sminthopsis macroura*	Monash	21.1	0.46	0.076	0.106	0.112	0.029	0.028	0.016	0.168	0.060	2.24	0.471	3.31
12	Fox, red	*Vulpes vulpes*	FSU	9,000	40	6.11	3.15	4.17	1.48	3.50	3.29	4.89	1.55	27.4	16.4	72.0
13	Glider, yellow-bellied	*Petaurus australis*	Monash	585	6.2	0.088	0.254	0.289	0.062	0.073	0.040	0.300	0.072	3.56	1.44	6.18
14	Glider, feathertailed	*Acrobates pygmaeus*	Monash	14.4	0.38	0.032	0.077	1.12	0.038	0.022	0.011	0.123	0.022	1.08	0.285	2.81
15	Glider, greater	*Petauroides volans*	Monash	1,350	4.14	0.132	0.251	0.522	0.106	0.101	0.047	0.566	0.149	8.29	5.24	15.4
16	Glider, sugar	*Petaurus breviceps*	Monash	130	2.69	0.160	0.460	0.551	0.113	0.043	0.044	0.195	0.134	3.63	2.59	7.92
17	Gopher, pocket	*Geomys bursarius*	FSU	175	1.35	0.545	0.311	0.333	0.098	0.231	0.040	0.188	0.132	1.07	0.260	3.20
18	Guinea pig	*Cavia porcellus*	FSU	415	4.6	0.63	1.83	3.06	0.28	0.33	0.22	1.10	0.16	7.17	4.73	19.51
19	Hedgehog	*Erinaceus europaeus*	FSU	372	2.26	0.743	0.647	0.317	0.163	0.223	0.001	0.700	0.195	4.36	3.65	11.0
20	Human	*Homo sapiens*	Afip.	64,922	1,240	16.2	19.7	9.72	1.51	3.22	2.60	7.30	3.81	65.2	58.0	187
21	Kangaroo rat	*Dipodomys merriami*	FSU	35.0	0.39	0.555	0.559	0.165	0.219	0.169	0.159	0.256	0.183	1.88	0.485	4.63
22	Kangaroo, eastern grey	*Macropus giganteus*	Monash	82,000	73.3	0.660	1.71	2.43	0.320	0.634	0.194	2.12	0.619	22.5	22.1	53.3
23	Lemur, brown	*Petterus fulvus*	FSU	2,000	9.50	3.38	1.60	0.704	0.542	1.31	0.573	1.56	0.383	7.20	6.21	23.5
24	Llama	*Llama peruana*	U. Wisc.	60,000	200.3	4.28	3.05	8.38	1.62	2.00	1.92	4.51	0.356	51.8	32.6	111
25	Loris, slow	*Nycticebus coucang*	FSU	965	7.88	1.31	0.933	0.363	0.271	0.600	0.298	0.639	0.170	4.80	2.97	12.4
26	Marsupial mouse, dusky	*Antechinus swainsonii*	Monash	18	0.59	0.101	0.121	0.205	0.050	0.058	0.038	0.029	0.033	1.37	0.316	2.32
27	Mink	*Mustela vison*	U. Wisc.	1,495		0.658	0.517	0.950	0.164	0.200	0.160	0.483	0.250	5.31	4.04	12.7
28	Mole	*Scalopus aquaticus*	FSU	53.0	0.57	0.080	0.100	0.050	0.040	0.038	0.017	0.145	0.030	0.405	0.435	1.34
29	Mole rat	*Nannospalax ehrenbergi*	FSU	50.0	0.51	0.387	0.194	0.131	0.088	0.097	0.037	0.171	0.040	0.877	1.03	3.05
30	Monkey, green	*Cercopithecus aethiops*	FSU	1,370	36.54	0.650	1.02	0.671	0.203	0.474	0.313	1.06	0.281	4.03	6.67	15.4
31	Monkey, macaque	*Macaca mulatta*	FSU	3,300	53.70	1.34	1.45	0.605	0.239	0.545	0.301	1.36	0.799	6.86	8.67	22.2
32	Mountain beaver	*Aplodotia rufa*	U. Wisc.	680	5.86	0.415	0.603	2.92	0.123	0.174	0.114	0.954	0.207	5.60	4.86	16.0
33	Mouse	*Mus musculus*	FSU	27	0.40	0.365	0.155	0.233	0.096	0.320	0.001	0.150	0.042	0.700	0.240	2.28
34	Mouse, spiny	*Acomys cahirinus*	RH-FSU	44	0.76	0.21	0.14	0.21	0.28	0.28	0.02	0.33	0.09	2.25	0.65	3.99
35	Opossum, didelphis	*Didelphis virginiana*	FSU	2,727	4.36	0.95	1.28	1.36	0.180	0.290	0.110	1.00	0.415	3.49	0.875	11.8
36	Opossum, marmosa	*Marmosa elegans*	FSU	54	0.51	0.235	0.265	0.320	0.105	0.120	0.060	0.330	0.045	4.80	0.565	6.85
37	Opossum, monodelphis	*Monodelphis domestica*	FSU	83.0	0.79	0.410	0.480	0.410	0.135	0.085	0.020	0.325	0.155	4.49	0.690	7.20
38	Possum, honey	*Tarsipes rostratus*	Monash	14.2	0.33	0.060	0.110	0.128	0.043	0.053	0.023	0.132	0.051	0.970	0.213	1.78
39	Possum, leadbeaters	*Gymnobelideus leadbeateri*	Monash	109.5	2.38	0.095	0.320	0.356	0.111	0.086	0.052	0.260	0.080	3.19	0.885	5.44
40	Possum, ring-tail	*Pseudocheirus peregrinus*	Monash	840	5.43	0.297	0.631	0.645	0.204	0.168	0.081	0.443	0.125	5.38	2.01	9.99
41	Possum, striped	*Dactylopsila trivirgata*	Monash	410	6.85	0.173	0.330	0.436	0.125	0.082	0.058	0.568	0.191	5.79	2.16	9.91
42	Pouched mouse, jumping	*Antechinomys laniger*	Monash	22.9	0.59	0.072	0.133	0.135	0.076	0.048	0.011	0.180	0.060	1.40	0.210	2.33
43	Prairie dog	*Cynomys ludovicianus*	FSU	1,250	4.96	0.53	0.75	1.10	0.31	0.28	0.15	1.37	0.47	8.27	3.21	16.44
44	Quoll	*Dasyurus hallucatus*	Monash	(550)	2.83	0.455	0.475	0.830	0.155	0.450	0.115	1.05	0.485	10.5	2.26	16.8
45	Rabbit	*Oryctolagus cuniculus*	FSU	5,230	7.57	1.34	0.943	1.43	0.235	0.296	0.197	0.760	0.216	5.56	2.78	13.8
46	Raccoon	*Procyon lotor*	FSU	4,309	27.4	3.28	2.28	1.79	0.572	1.36	0.434	1.44	0.509	7.59	5.23	24.5
47	Rat, albino	*Rattus rattus*	FSU	268	1.50	0.130	0.950	0.765	0.305	0.200	0.075	0.495	0.145	5.70	1.83	10.6
48	Rat, Norway	*Rattus norvegicus*	RH-FSU	226	2.17	0.92	0.81	1.03	0.69	0.77	0.16	3.15	0.95	11.3	2.16	21.94
49	Sheep	*Ovis aries*	FSU	32,950	125.9	2.12	1.83	3.36	0.577	1.19	0.636	2.51	0.822	21.1	18.8	52.9
50	Squirrel	*Sciurius carolinensis*	FSU	367	6.49	0.600	0.630	1.17	0.280	0.210	0.185	1.33	0.520	8.10	2.85	15.9
51	Tree shrew	*Tupaia glis*	FSU	145	2.02	0.24	0.75	0.94	0.20	0.06	0.09	0.44	0.17	3.93	2.55	9.37
52	Wallaby, Little Rock	*Perodorcas concinna*	Monash	1,020	8.39	0.230	1.26	1.40	0.125	0.170	0.080	0.571	0.168	10.1	3.78	17.9
53	Wallaby, swamp	*Wallabia bicolor*	Monash	10,000	34.36	0.735	1.76	4.61	0.450	0.710	0.265	2.00	0.545	28.9	12.7	52.6

Body weights in parentheses are estimates from Nowak [1991], others weighed at time of dissection. Estimate of body and brain weight of little brown bat courtesy of Dr. Donald Wong, Indiana University.

AVCN = Anteroventral cochlear nucleus; PVCN = posteroventral cochlear nucleus; DCN = dorsal cochlear nucleus; MTB = medial nucleus of the trapezoid body; LSO = lateral superior olive; MSO = medial superior olive; VLL = ventral nucleus of lateral lemniscus; DLL = dorsal nucleus of lateral lemniscus; IC = inferior colliculus; MG = medial geniculate.

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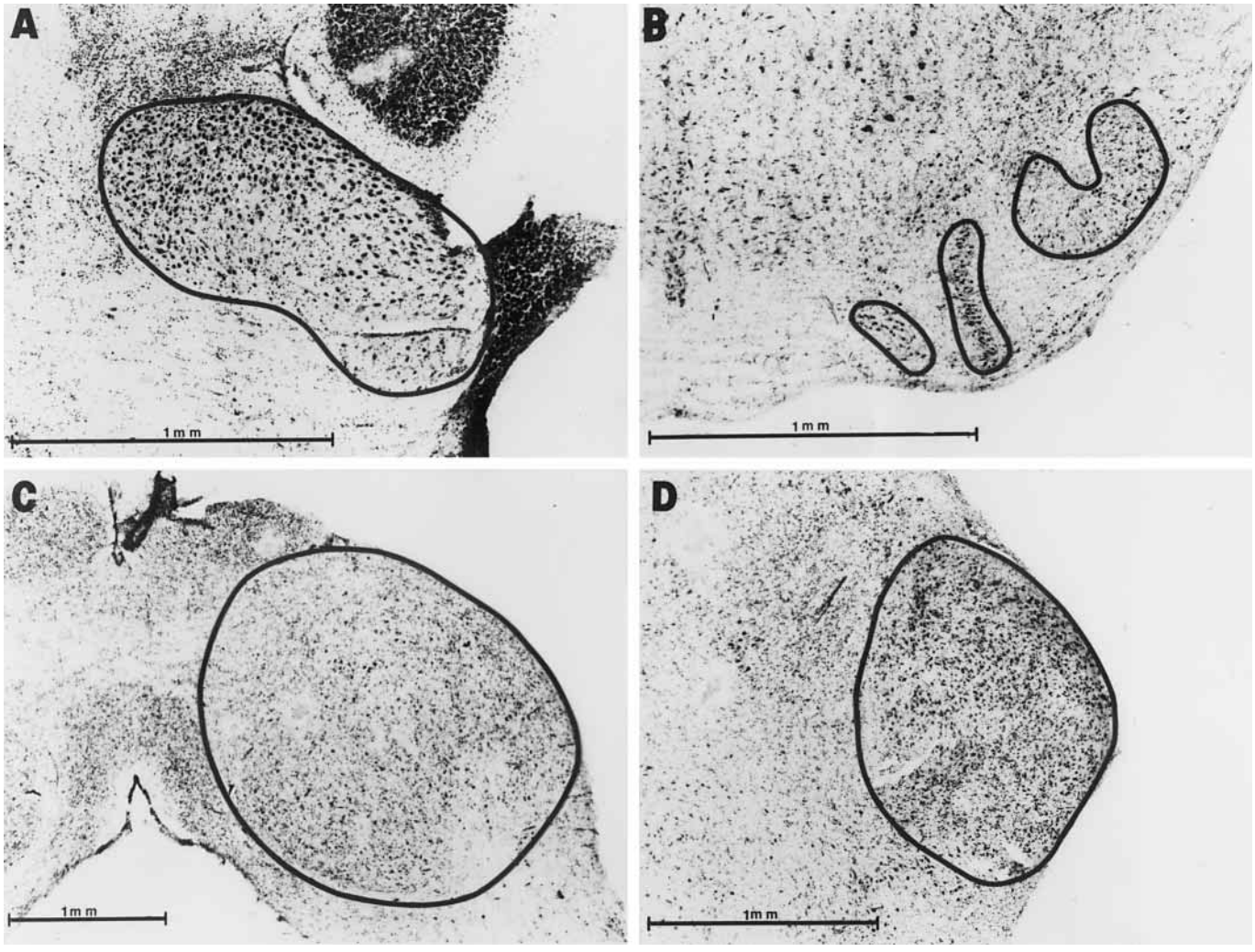


Fig. 1. Photomicrographs of four Nissl-stained transverse sections through six auditory nuclei in a kangaroo rat. **A** Anteroventral cochlear nucleus; **B** Superior olivary complex; **C** Inferior colliculus; **D** Medial geniculate. The encircling lines indicate the borders drawn for morphometric measurement. Measurement bars indicate actual size in mm. Note that a single 'nucleus' might include several subdivisions, many distinguishable cell-types and an arbitrary amount of neuropil and fibers. Despite this lack of morphological refinement, random measurement error proved to be less than 3% within each animal in the sample (see fig. 2).

fused with formalin and embedded in celloidin but 16 brains were frozen-sectioned after formalin or glutaraldehyde perfusion and the remaining 6 were paraffin-embedded. Each of these histological techniques is known to shrink neurological materials. For example, perfusion, dehydration, and either celloidin or paraffin embedding before sectioning can shrink a brain more than 20% [e.g., Berman, 1968]. Similarly, formalin perfusion and then formalin-sucrose soaking and frozen sectioning can shrink materials about 7% [e.g., Pellegrino et al., 1979].

Furthermore, the shrinkage from each of these histological techniques may not be uniform. Surface structures (DCN, IC, MG) may be more shrunken than deep structures (e.g., LSO, MSO, or MTB).

Therefore, the in vitro sections used for volume measurements here are *not* valid representations of the same nuclear volumes in vivo nor even of the same nuclear volumes in sections treated in a different histological manner. Because the shrinkage of the present material during histological handling was undocumented if measured at all, an unmeasured bias in the raw data is certainly present. Of particular importance here, this shrinkage varies among the animals, and hence it is confounded with the natural variation among the same animals if no countermeasure is taken.

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This bias due to differential shrinkage is a formidable problem for conventional allometry studies where brain volume or nuclear volumes are related to body weight or to brain weight, because the brain and its constituent nuclei shrink in volume (and gain in weight) with histological processing whereas body weight is usually measured before sacrifice and remains unbiased. In such studies, painstaking corrections of the raw data are required before an interpretable analysis can proceed [e.g., Stephan and Andy, 1969; Stephan et al., 1991; and see Johnson and Kirsch, 1993].

It is important to note, however, that for more modest studies, like the present one, the error in the measurement of raw nuclear volumes due to relatively *uniform* shrinkage can be removed by simple arithmetical procedures. It is minimized, if not removed, in the present study by restricting each comparison to *relative* instead of absolute (raw) volumes. For example, in the first of the two types of comparison, the measurements of raw volumes are first transformed into relative volumes or percentages to standardize them on a common basis. Such proportions or percentages are unit-less numbers, and hence they are *not* subject to a bias due to uniform shrinkage nor to any other near constant bias. However, the use of these relative sizes instead of absolute sizes means that relationships between the nuclei and other non-nuclear measures such as body size are obviated. In the other type of comparison, least-squares correlation coefficients [Pearson 'r'; Senter, 1969] are used to judge relative 'similarity' among the animals. A correlation coefficient, 'r', unlike its parent regression equation, is also a unit-less number. Hence, correlations and their derivatives (e.g., r^2, $1–r^2$) either between animals or between nuclei within animals are not altered by a uniform bias in the raw data regardless of the amount of the bias. For this reason, the comparisons among the animals or among the nuclei within an animal have been restricted to correlation coefficients (used as 'similarity indices'), avoiding regression analysis entirely. Indeed, in the 'scatterplots' shown in the figures to follow, the slope of the 'regression line' and the axis scaling are meaningless, since they change with degree of shrinkage. Nevertheless, the least-squares lines are shown so the 'goodness of fit' (i.e., the correlation coefficient) can be seen and judged.

Therefore, any *uniform* shrinkage among the nuclei within an animal, and the differences in shrinkage due to differences in histological processing among the animals, probably have little effect on the chief results, statistics and conclusions.

Insofar as there may be *non*-uniform shrinkage among the nuclei within an animal, however, such non-uniformity remains a source of uncontrolled and unavoidable error. It has not and cannot be eliminated in the present data. Therefore, all conclusions based on the present materials must be weakened accordingly.

Single Animal Error

Another source of potential error in morphometry studies of many species is the use of a single individual to represent an entire species. Often the relative age and gender of the individuals are undocumented. This study is no exception. The sectioned brains were almost certainly from 'adults', but 10 were females and 43 were males.

Although this uncertainty in the data could be disastrous for some kinds of comparative studies, the constraint on the measurements already discussed – namely, the use of only relative measures and unit-less statistics – would seem to minimize this potential problem too. For example, a small female and a large male are presumably little different in the *relative* size of their auditory nuclei even though the absolute size of their nuclei (and brains and bodies) are often quite different. In general, only if dimorphic differences resulted in differences in the *relative* size of the auditory nuclei would the present analysis be compromised. Although such differences are known to be present in some non-mammals [e.g., Nottebohm et al., 1994], we know of no evidence that they are present among the central auditory nuclei in mammals. Nevertheless, this potential source of uncontrolled error also serves to weaken the certainty of the results and conclusions.

Drawing Error

The variation inherent in judging the borders and then drawing the outline and measuring the area of specific nuclei in section after section through each animal's brain is a source both of random error and of experimenter bias. We measured both of these types of error before proceeding.

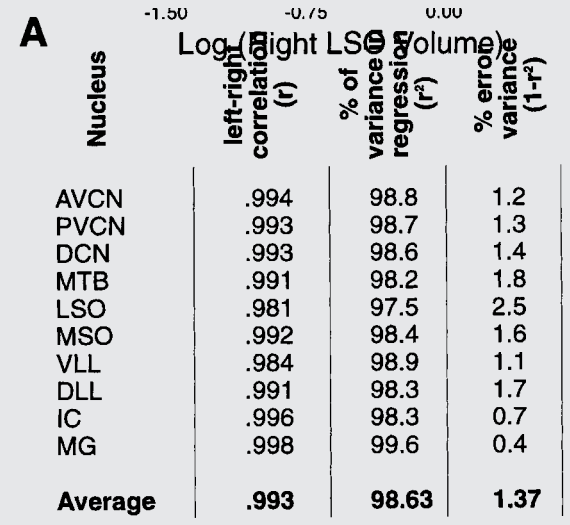
To begin with, the reliability or random component of these errors can be measured by comparing the estimate of the size of each nucleus on one side of the brain with the estimate of the size of its contralateral counterpart: that is, by deriving the error between each left-right pair of nuclei among the 53 animals. One way this error can be estimated for the entire set of measurements while maintaining unit-less measures, is by deriving the 10 correlation coefficients

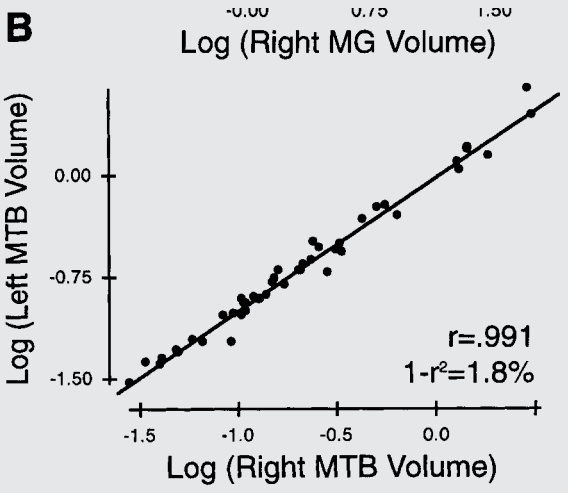
Fig. 2. Error variance between each nucleus and its contralateral counterpart. **A** Percentage error variance for each nucleus among 47 animals (df = 45). **B**–**F** Representative scatterplots showing derivation of 5 of the values in **A**.

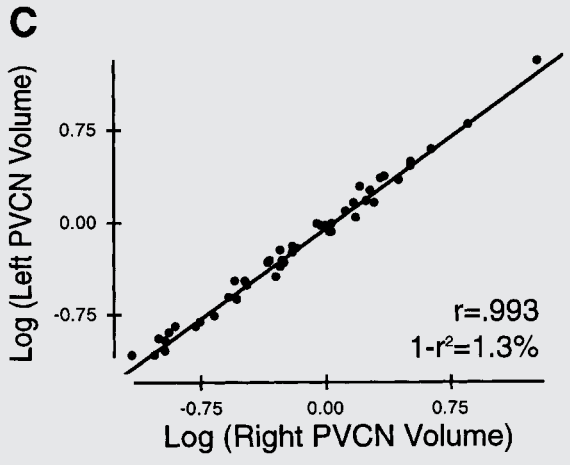
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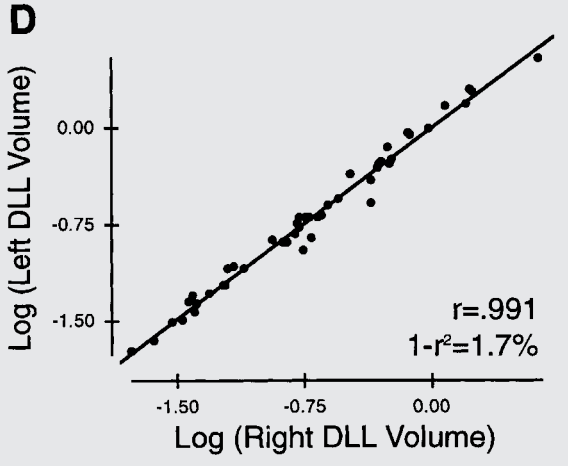
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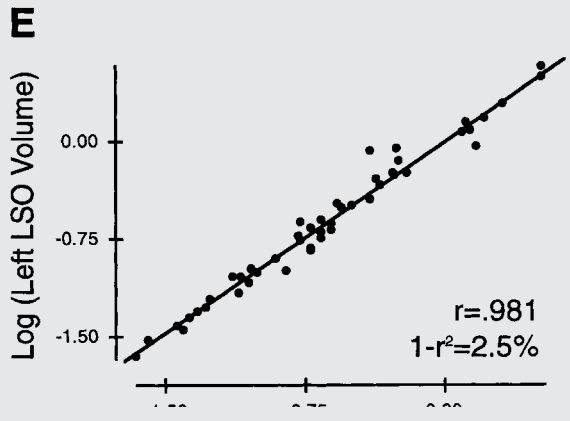
Nucleus	left-right correlation (r)	% of variance in regression (r^2)	% error variance ($1-r^2$)
AVCN	.994	98.8	1.2
PVCN	.993	98.7	1.3
DCN	.993	98.6	1.4
MTB	.991	98.2	1.8
LSO	.981	97.5	2.5
MSO	.992	98.4	1.6
VLL	.984	98.9	1.1
DLL	.991	98.3	1.7
IC	.996	98.3	0.7
MG	.998	99.6	0.4
Average	**.993**	**98.63**	**1.37**

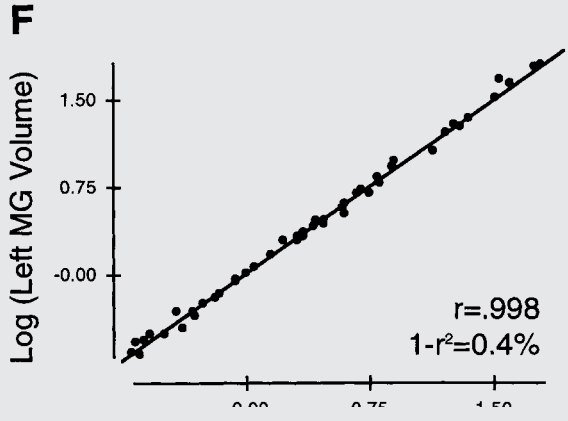












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between each nucleus and its contralateral counterpart for the entire set of animals. One outcome of this computation, besides the correlation itself, is the so-called 'residual variance' – the variation *not* accounted for by the correlation with a regression line whose slope = 1. Therefore, average 'residual variation' can be used as an estimate of the outlining 'error' in the 530 left-right pairs of nuclei (10 pairs of nuclei × 53 animals). It can be noted that if there was no error at all, each correlation would be 1.0 and the residual or error variance would be 0. As one or another of the 10 correlations lessens from 1.0 toward 0, however, the error or residual variance increases from 0 to 1.0 according to the equation: residual or error variance = $1 - r^2$.

These left-right error calculations, using log volume measurements to normalize the data, are shown in figure 2. Figure 2 also contains five representative scatterplots showing the source of 5 of the 10 entries in the table (table 1). As can be seen, the error ranged from a high of 2.5% (for LSO, fig. 2E) to a low of 0.4% (for MG, fig. 2F). The average error among the 10 nuclei was 1.37% on the transformed (log) scale shown and 2.9% on the original percentage scale.

The arbitrary judgements of the nuclear borders by the experimenters is a second and independent source of uncontrolled drawing error. There is little question that different observers would have drawn the nuclear borders somewhat differently, including more or less neuropil or more or less embedded or bypassing fibers, etc. (fig. 1). This source of bias during outlining obviously affects the raw volume measurements. However, if the bias is relatively constant for each drawer, then for the same reason as discussed above for shrinkage, this bias is removed by restricting the analysis to unit-less numbers – relative sizes and correlation coefficients. Therefore, this source of bias probably has little effect on the standardized or relative size measurements, or the correlations, or the conclusions based on them. In this case, the total 'drawing and area-measurement error' is reduced to the non-reliability of the measurements, and this source of error as just shown, averages about 3% for the relative volumes.

In summary, the outlining error – its bias and its unreliability – whether the result of inconsistent judgement or inconsistent or inaccurate drawing or planimetry, averages less than 3% among the 10 nuclei. Experimenter bias, contained within this error estimate, is mostly removed by the restriction of interspecies comparisons to unit-less numbers.

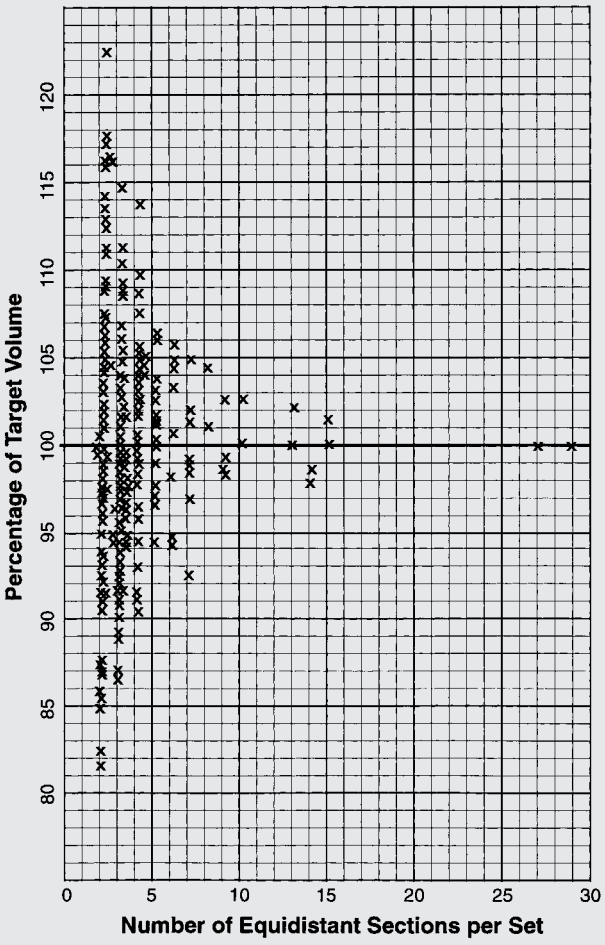


Fig. 3. Empirical verification of accuracy of estimate of the volume of the cochlear nucleus in cats as a function of the number of sections in the sample set (courtesy of Dr. J. Brunso-Bechtold). Note convergence of estimates toward target volume. Although error of 3% is probably attained with 9 or 10 equidistant sections, this number was effectively doubled (and error made still smaller) by averaging volume estimates of the left and right nuclei (table 1).

Error in Approximating Nuclear Volumes from Incomplete 'Stacks' of Sections

The fourth type of error peculiar to the present technique obtains with the approximation of the volume of a nucleus by means of summing its cross-sectional areas in separate sections. Because the thickness of the sections varied from 8 μm for some of the small animals to 50 μm for the larger animals (human, deer, etc.), there were often dozens of sections through a large nucleus (IC, MG, DCN, etc.) – a number too large for the task of outlining and mea-

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suring to remain bearable. Therefore, we routinely drew and measured a systematic sample of the sections with a fixed but arbitrary inter-section distance (see Methods). The question arose as to the number of sections that should be included in the sample of sections through a nucleus in order to reduce the error in approximation to an acceptable level.

Although this type of error can be calculated theoretically [e.g., Boole, 1958], we derived an estimate empirically, using sections through the cochlear nucleus of a cat. We began by measuring the area of left and right DCN and VCN (AVCN + PVCN) in over 200 50 μm-sections and, as described in the Methods, multiplying each area by the inter-section distance (along with the several necessary measurement unit and magnification corrections). This measurement was taken as the 'target' volume which was to be approximated with the lesser sets of sections selected for the experiment.

To establish the number of sections necessary for a suitable approximation, we subdivided the entire set of sections into systematic subsamples of equidistant sections, ranging from only 2 sections per set through 3, 4, 5, … to 29 sections per set, in each case recalculating the estimated nuclear volume anew. In all, we measured 65 sets with 2 sections per set, 65 sets with 3 sections per set, 26 sets with 4 sections per set, etc.

Figure 3 shows the outcome of these successive measurements in terms of their percentage approximation of the target set. It can be seen in the figure that the many estimates of the nuclear volume converged rapidly toward that of the target volume as a function of the number of sections per set. For example, with 2 sections per set the 65 sample sets ranged from about 80 to 122% of the target volume (i.e., about ±25%), with 3 sections per set, the 65 sets ranged from 86 to 115% (i.e., about ±15%), with 4 sections from 90 to 114%, with 5 sections from 94 to 107%, etc.

For the present purpose the useful result was that with only 5 equidistant sections per set, the range of variation was within 10% of the target volume; with 8 sections per set, the range of variation was within 5%; with 9 sections, it was within 3%; with 15 sections, within 2%; and with 27 sections per set, the error was less than 0.1%. When the same series of repeated measurements was performed with DCN alone, a still quicker convergence on the target volume was obtained – probably the result of DCN's more smoothly convex border.

From these data, together with our experience with the laboriousness of the task, we chose, arbitrarily, a 3% error as sufficiently accurate for the present purpose. Henceforth, we strove to include at least 9 sections through each nucleus for each animal. This number of sections per set was easily attained in the vast majority of animals and for all of the large nuclei (AVCN, PVCN, DCN, VLL, IC, and MG). Only the smaller nuclei (e.g., MSO, DLL) in the smallest animals (e.g., mole, honey possum) did not immediately achieve this criterion. Therefore, we adopted yet another technique to reduce the error still further – we averaged the volume of the left and right nuclei. Because in no case were the histological sections bilaterally symmetrical, this averaging, in effect, doubled the number of sections in the set estimating the nuclear volume and hence reduced the probable error to about 2%.

Summary of Measurement Error Inherent in the Technique

From these preliminary considerations and experiments on the technique itself, it is probably safe to conclude that the *random error* inherent in the raw measures of in vitro volume is less than 5% and probably less than 3% for most measurements. As will be seen in the descriptive graphs that follow, this amount of error, when placed within the more than 100-fold range of nuclear sizes among the animals, proves to be less than the size of the symbols used to depict the results – rendering 'error bars' unnecessary.

Morphometry Results

Turning to the chief results, table 1 shows the primary (or raw) morphometry data. The entries in the table are the in vitro volumes (in mm^3) of 10 auditory nuclei in each of the 53 animals. In addition, Column 5 shows the body weight and Column 6 the brain weight for each animal for which these data were recorded at time of sacrifice and dissection. Finally, the last Column ('total') shows the (calculated) sums of the volumes of the 10 individual nuclei for each animal. These totals were used later to derive the relative volumes of the nuclei in percentages of the total subcortical system, hence standardizing the entries for inter-animal comparison. All further tables, figures, statistics and conclusions are derived from the entries in this table after appropriate standardizing and/or normalizing transformations.

Variation in Relative Size of the Auditory System

The *absolute* size of the entire subcortical auditory system varies more than 139-fold among the 53 animals in the sample (see mole vs. human data in last Column, table 1). Although this amount of variation might seem very large, it is quite small compared to the 3700-fold range of variation

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in brain weights (see honey possum vs. human data in table 1) and the 48,000-fold variation in body weights (see bat vs. deer data in table 1). This range of variation in size among the animals in the sample is quite comparable to that obtained from still larger samples of mammals [e.g., see Hofman, 1982]. The variation in auditory system size (which *does* contain the added error variation due to different histological processing and gender differences) is shown graphically in figure 4.

Using the conventional log (raw volume) transformation to normalize the distribution, figure 4A shows that despite the histological error in the raw data, the total size of the ascending auditory system remains related to brain size ($r = 0.903$; $p < 0.0001$). Since the volumes of the auditory nuclei were shrunken during histology whereas brain weights were not, the slope of the regression line is almost meaningless. Nevertheless, figure 4 shows that despite the extra variation due to histological shrinkage, the size of the total subcortical auditory system remains related to brain size with some notable deviations.

Figure 4B shows the distribution of the sizes of the total subcortical auditory system relative to brain weight – that is, log (total auditory system volume/brain weight) – the deviations from the relationship shown in figure 4A. From this histogram (fig. 4B) it can be seen that bats, and then kangaroo rats, marmosa opossums and Norway rats have the largest auditory system relative to their brain size.

Perhaps as interesting, humans have the smallest auditory system per brain size. This very small ratio for humans is the passive result of a very large brain (the fraction's denominator) rather than a very small auditory system (the fraction's numerator). That is, humans have the largest auditory system (187 mm^3) *and* the largest brain size (1,240 gms). However, the auditory system expansion falls short of brain expansion yielding a very small auditory system/brain size ratio. That this small ratio is the secondary result of the expansion of other, non-auditory, brain parts is supported by noting that each of the six primates in the sample, all highly visual with expanded occipital and temporal lobes in their cerebral hemispheres, have an auditory system/brain size ratio smaller than the sample average ($p = 1/2^6 = 1/64 = 0.016$), and further, the ratio becomes progressively smaller over the 3-step sequence among the three families of primates: prosimians (bushbabys, lorises, lemurs), anthropoids (macaques and green monkeys), hominoids (humans). However, Johnson and Kirsch [1993], using essentially the same methods used here, have shown that non-cortical auditory and visual nuclei are unrelated, not inversely related. Therefore, deviations above the regression line in figure 4A indicative of a high auditory sys-

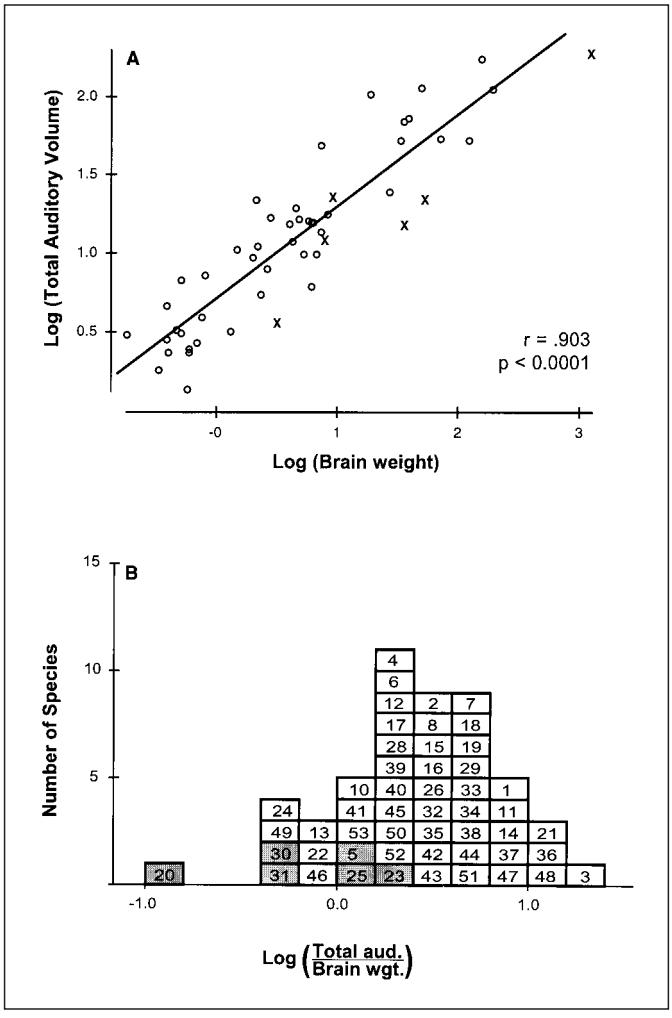


Fig. 4. Relationship of (log) volume of subcortical auditory system to (log) brain size. **A** Scatterplot of log volume of total brainstem auditory system (i.e., sum of volumes of 10 nuclei) versus log brain weight. Each point represents one animal. Although the slope of the least-squares line shown is meaningless because of the large and uncontrolled error in raw volume data due to differential histological shrinkage of nuclei, the size of the auditory system remains reliably related to brain size, x = primates. **B** Distribution of log auditory system size relative to log brain weight. Numbers refer to animal listing in table 1. Note that with this measure the little brown bat (n = 3) is largest; the human (n = 20) is smallest by far. X's in **A** and shaded boxes in **B** are primates. Five of the six primates in sample are below regression line in **A** and below mean in **B** – indicating unusually small auditory systems given brain size – probably the result of a radically expanded vision-related cortex.

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tem/brain weight ratio are properly interpreted as unusually large auditory systems, whereas deviations below the regression line are not so easily interpreted because they are often the result of the expansion of other, non-auditory, structures included here in the brain weight measurement and not in the auditory system measurement, such as the large cerebral hemispheres in the primates.

More refined analysis of this allometric relation between auditory system and brain size probably requires more histological constancy than afforded by the present materials.

Relative Sizes of Nuclei

In order to compare the forms of the animals' auditory systems despite their wide variation in absolute size, the array of nuclear volumes was first standardized for each animal. Since we know of no brain part, nucleus or tract that remains in a constant relationship with either body or brain size among all mammals, we chose as the most conservative standard the total size of the subcortical auditory system in each animal (i.e., see 'total' column in table 1). By dividing each of the 10 measured volumes for that animal by the sum of the 10 volumes (i.e., Columns 7 to 16, divided by Column 17, table 1), the raw volumes for each nucleus in each animal (with units of mm^3) were transformed to unit-less percentages of the volume of its entire system. These standardized data, shown as entries in table 2, are the sizes of each nucleus within each animal *relative* to the size of its total subcortical auditory system.

Using the human data (from Row 20, table1) to illustrate the effect of this standardizing transformation, table 2 (Row 20) shows that AVCN is about 8.7% of the total system, PVCN is 10.5%, DCN is 5.2%, IC is about 35%, and MG is 31% of the total auditory system in humans, etc.

Before turning to the animal-by-animal comparisons afforded by the data in table 2, it can be noted that these relative data show several inter-nuclear relationships that seem to affirm their face-value. For example, since the superior olives (LSO, MSO, MTB) receive most of their excitatory afferent supply from AVCN and probably none at all from DCN, it is satisfying to note that the size of the superior olives collectively (i.e., MTB+LSO+MSO or the sum of the 6th, 7th, and 8th columns in table 2) is reliably related to the size of AVCN ($r = 0.744$, $p < 0.001$). At the same time, these nuclei are *not* related to the size of DCN ($r = -0.142$). Similarly, the size of LSO is reliably related to the size of MTB ($r = 0.515$, $p < 0.01$), the source of most of LSO's inhibitory input [e.g., Glendenning et al., 1991]. Therefore, despite the loss of detail resulting from the use of relative or percentage data, the data in table 2 retain a veridical description of each animal's auditory system.

Animals with Deviant Nuclei

Turning now to the distribution of sizes for each nucleus alone, table 3 shows the data in table 2 after logarithmic transformation of the raw data (to remove the skew) and then normalization of these log data to a mean of 0 and a standard deviation of 1 by subtracting the calculated mean and dividing by the standard deviation. Therefore, the entries for each nucleus (or nuclear cluster) in table 3 have an average of 0, a standard deviation of 1, and a S.E.M. of $1/\sqrt{53}$.

Table 4 uses the data in table 3 to obtain lists of the relative size among the 53 animals for each of the 10 nuclei, along with two nuclear clusters – the cochlear nucleus (CN = AVCN + PVCN + DCN) and the superior olives (SO = MTB + LSO + MSO). The 12 lists in table 4 are arranged so that the animal with the largest nucleus (or nuclear cluster) is at the top, while the animal with the smallest nucleus is at the bottom. The average size of each nucleus is also indicated in each list, and the statistically reliable deviants after the logarithmic transformation and normalization in table 3 are designated with asterisks in table 4.

For example, in the rankings in table 4A, the DCN's of the feathertailed gliders and the mountain beavers can be seen to be the largest among the 53 animals (see top of Column 3, 'DCN', in table 4A) and the asterisks show that in either animal DCN is far larger than is expected on the basis of DCN's distribution within the entire sample. We return to these two cases below.

The table also shows animals whose nuclei are reliably smaller than average (at the bottom of each column in table 4). Particularly notable among these entries are two found at the bottom of the MSO column, mice and hedgehogs. These two animals do not seem to have an MSO at all [e.g., table 1; Masterton et al., 1975]. Interesting in its own right, MSO is the only nucleus among the 10 that was found to be entirely absent in any of the animals in the sample. (Since log [0] is undefined, these animals were arbitrarily assigned log [0.001] so the normalization in table 3 was not obviated.)

In the past, some of the deviants in the lists in table 4 have been used to gain insight into the possible contributions of the nuclei [e.g., see Merzenich et al., 1973; Masterton et al., 1975; Aitkin and Nelson, 1989]. However, a direct correspondence between a deviant nucleus and its host's *current* ecological niche is usually disappointing and sometimes misleading. One of the reasons for the absence of such structure-ecology correspondences is due to the effect of structural adaptations to the special ecologies of the

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Table 2. Percentage size of each nucleus relative to the size of the total subcortical auditory system for each animal

No.	Animal	AVCN	PVCN	DCN	MTB	LSO	MSO	VLL	DLL	IC	MG
1	Armadillo	7.54	4.99	7.78	1.17	1.12	0.509	2.26	2.00	56.2	16.6
2	Bandicoot	3.14	5.91	7.48	1.89	1.51	0.629	6.13	0.975	59.1	13.3
3	Bat, little brown	18.5	2.99	4.53	3.76	4.19	2.15	3.19	1.31	21.1	38.3
4	Beaver	3.77	1.40	5.24	1.77	0.549	0.708	4.54	0.830	53.3	27.9
5	Bushbaby	10.8	12.1	11.2	1.43	2.75	2.33	5.03	2.13	34.0	18.4
6	Capybara	4.27	1.13	5.90	2.51	1.06	0.785	3.67	1.57	42.5	36.4
7	Cat	3.11	6.45	5.29	1.21	1.50	1.69	3.62	1.28	36.5	39.3
8	Colocolo	2.98	6.23	10.5	2.23	1.43	1.09	4.04	0.679	56.6	14.3
9	Chinchilla	2.61	7.59	12.7	1.16	1.37	0.912	4.56	0.664	29.7	19.6
10	Deer	5.59	2.32	6.63	2.06	0.355	0.663	1.45	1.05	44.9	35.0
11	Dunnart, darling downs	2.30	3.20	3.38	0.876	0.846	0.483	5.08	1.81	67.7	14.2
12	Fox, red	8.49	4.38	5.79	2.06	4.86	4.57	6.79	2.15	38.1	22.8
13	Glider, yellow-bellied	1.42	4.11	4.68	1.00	1.18	0.647	4.85	1.17	57.6	23.3
14	Glider, feathertailed	1.14	2.74	39.9	1.35	0.783	0.391	4.38	0.783	38.4	10.1
15	Glider, greater	0.857	1.63	3.39	0.688	0.656	0.305	3.68	0.968	53.8	34.0
16	Glider, sugar	2.02	5.81	6.96	1.43	0.543	0.556	2.46	1.69	45.8	32.7
17	Gopher, pocket	17.0	9.72	10.4	3.06	7.22	1.25	5.88	4.12	33.4	8.12
18	Guinea pig	3.23	9.38	15.7	1.44	1.69	1.13	5.64	0.820	36.8	24.2
19	Hedgehog	6.75	5.88	2.88	1.48	2.03	0.009	6.36	1.77	39.6	33.2
20	Human	8.66	10.5	5.20	0.807	1.72	1.39	3.90	2.04	34.9	31.0
21	Kangaroo rat	12.0	12.1	3.56	4.73	3.65	3.43	5.53	3.95	40.6	10.5
22	Kangaroo, eastern grey	1.24	3.21	4.56	0.600	1.19	0.364	3.98	1.16	42.2	41.5
23	Lemur, brown	14.4	6.81	3.00	2.31	5.57	2.44	6.64	1.63	30.6	26.4
24	Llama	3.86	2.75	7.55	1.46	1.80	1.73	4.06	0.321	46.7	29.4
25	Loris, slow	10.6	7.52	2.93	2.19	4.84	2.40	5.15	1.37	38.7	24.0
26	Marsupial mouse, dusky	4.35	5.22	8.84	2.16	2.50	1.64	1.25	1.42	59.1	13.6
27	Mink	5.18	4.07	7.48	1.29	1.57	1.26	3.80	1.97	41.8	31.8
28	Mole	5.97	7.46	3.73	2.99	2.84	1.27	10.8	2.24	30.2	32.5
29	Mole rat	12.7	6.36	4.30	2.89	3.18	1.21	5.61	1.31	28.8	33.8
30	Monkey, green	4.22	6.62	4.36	1.32	3.08	2.03	6.88	1.82	26.2	43.3
31	Monkey, macaque	6.04	6.53	2.73	1.08	2.45	1.36	6.13	3.60	30.9	39.1
32	Mountain beaver	2.59	3.77	18.2	0.769	1.09	0.713	5.96	1.29	35.0	30.4
33	Mouse	16.0	6.80	10.2	4.21	14.0	0.044	6.58	1.84	30.7	10.5
34	Mouse, spiny	5.26	3.51	5.26	7.02	7.02	0.501	8.27	2.26	56.4	16.3
35	Opossum, didelphis	8.05	10.8	11.5	1.53	2.46	0.932	8.47	3.52	29.6	7.42
36	Opossum, marmosa	3.43	3.87	4.67	1.53	1.75	0.876	4.82	0.657	70.1	8.25
37	Opossum, monodelphis	5.69	6.67	5.69	1.88	1.18	0.278	4.51	2.15	62.4	9.58
38	Possum, honey	3.37	6.18	7.19	2.42	2.98	1.29	7.42	2.87	54.5	12.0
39	Possum, leadbeaters	1.75	5.88	6.54	2.04	1.58	0.956	4.78	1.47	58.6	16.3
40	Possum, ring-tail	2.97	6.32	6.46	2.04	1.68	0.811	4.43	1.25	53.9	20.1
41	Possum, striped	1.75	3.33	4.40	1.26	0.827	0.585	5.73	1.93	58.4	21.8
42	Pouched mouse, jumping	3.09	5.71	5.79	3.26	2.06	0.472	7.73	2.58	60.1	9.01
43	Prairie dog	3.22	4.56	6.69	1.89	1.70	0.912	8.33	2.86	50.3	19.5
44	Quoll	2.71	2.83	4.94	0.923	2.68	0.685	6.25	2.89	62.5	13.5
45	Rabbit	9.71	6.83	10.4	1.70	2.14	1.43	5.51	1.57	40.3	20.1
46	Raccoon	13.4	9.31	7.31	2.33	5.55	1.77	5.88	2.08	31.0	21.3
47	Rat, albino	1.23	8.96	7.22	2.88	1.89	0.708	4.67	1.37	53.8	17.3
48	Rat, Norway	4.19	3.69	4.69	3.14	3.51	0.729	14.4	4.33	51.5	9.85
49	Sheep	4.01	3.46	6.35	1.09	2.25	1.20	4.74	1.55	39.9	35.5
50	Squirrel	3.77	3.96	7.36	1.76	1.32	1.16	8.36	3.27	50.9	17.9
51	Tree shrew	2.56	8.00	10.0	2.13	0.640	0.961	4.70	1.81	41.9	27.2
52	Wallaby, Little Rock	1.28	7.04	7.82	0.698	0.950	0.447	3.19	0.939	56.4	21.1
53	Wallaby, swamp	1.40	3.35	8.76	0.856	1.35	0.504	3.80	1.04	54.9	24.1
	Average	5.59	5.70	7.47	1.96	2.46	1.12	5.39	1.81	45.3	22.8
	Standard deviation	4.45	2.65	5.54	1.15	2.26	0.83	2.22	0.93	12.1	10.1
	% variation	0.80	0.46	0.74	0.59	0.92	0.74	0.41	0.51	0.27	0.44

Note that each row sums to 100%. Statistics at bottom of each column are logarithmic normalized values (i.e., antilogs of mean and standard deviation derived from logs of table entries). Percent variation is 100× standard deviation/mean.

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Table 3. Normalized logarithmic values of entries in table 2

No.	Animal	AVCN	PVCN	DCN	CN	MTB	LSO	MSO	SO	VLL	DLL	IC	MG
1	Armadillo	0.755	–0.004	0.375	0.391	–0.670	–0.721	–0.782	–1.00	–1.83	0.439	0.899	–0.420
2	Bandicoot	–0.381	0.308	0.297	–0.123	0.217	–0.296	–0.535	–0.336	0.461	–0.870	1.08	–0.870
3	Bat, little brown	1.91	–0.969	–0.716	1.01	1.55	1.11	1.16	1.40	–1.05	–0.374	–2.59	1.28
4	Beaver	–0.119	–2.41	–0.429	–1.27	0.112	–1.80	–0.327	–0.874	–0.253	–1.29	0.711	0.634
5	Bushbaby	1.22	1.66	1.09	1.68	–0.373	0.559	1.28	0.573	–0.006	0.531	–0.890	–0.211
6	Capybara	0.040	–2.86	–0.179	–1.06	0.746	–0.721	–0.146	–0.158	–0.712	0.018	–0.095	1.17
7	Cat	–0.381	0.491	–0.391	–0.376	–0.670	–0.296	0.874	–0.158	–0.777	–0.374	–0.637	1.33
8	Colocolo	–0.423	0.402	0.964	0.317	0.500	–0.391	0.285	–0.034	–0.529	–1.54	0.925	–0.723
9	Chinchilla	–0.607	0.786	1.34	0.690	–0.670	–0.391	0.013	–0.586	–0.201	–1.54	–1.37	–0.083
10	Deer	0.379	–1.47	0.044	–0.444	0.410	–2.11	–0.327	–0.754	–2.99	–0.870	0.100	1.09
11	Dunnart, darling downs	–0.764	–0.847	–1.27	–1.65	–1.23	–1.16	–0.782	–1.46	0.041	0.240	1.56	–0.737
12	Fox, red	0.915	–0.246	–0.213	0.188	0.410	1.33	2.22	1.66	0.716	0.619	–0.484	0.224
13	Glider, yellow-bellied	–1.40	–0.379	–0.630	–1.32	–1.02	–0.602	–0.535	–1.00	–0.053	–0.525	0.987	0.268
14	Glider, feathertailed	–1.71	–1.17	3.61	2.29	–0.373	–1.16	–1.08	–1.14	–0.306	–1.29	–0.456	–1.43
15	Glider, greater	–1.97	–2.16	–1.27	–2.68	–1.71	–1.34	–1.47	–1.94	–0.712	–0.870	0.744	1.03
16	Glider, sugar	–0.944	0.276	0.160	–0.393	–0.373	–1.80	–0.535	–1.22	–1.63	0.132	0.171	0.956
17	Gopher, pocket	1.81	1.25	0.945	1.89	1.16	1.85	0.402	1.64	0.383	1.79	–0.953	–1.88
18	Guinea pig	–0.340	1.19	1.76	1.22	–0.373	–0.125	0.285	–0.245	0.260	–1.29	–0.608	0.345
19	Hedgehog	0.629	0.308	–1.59	–0.262	–0.240	0.098	–2.96	–0.533	0.574	0.240	–0.347	0.987
20	Human	0.945	1.40	–0.429	0.848	–1.45	–0.125	0.611	–0.383	–0.589	0.439	–0.797	0.847
21	Kangaroo rat	1.36	1.66	–1.16	1.16	1.96	0.941	1.81	1.69	0.218	1.75	–0.258	–1.35
22	Kangaroo, eastern grey	–1.60	–0.847	–0.672	–1.63	–2.01	–0.602	–1.08	–1.46	–0.529	–0.525	–0.120	1.44
23	Lemur, brown	1.59	0.576	–1.52	0.828	0.585	1.51	1.34	1.43	0.646	0.018	–1.26	0.522
24	Llama	–0.086	–1.17	0.297	–0.513	–0.240	–0.046	0.874	0.082	–0.471	–3.14	0.240	0.740
25	Loris, slow	1.20	0.761	–1.59	0.476	0.500	1.30	1.34	1.26	0.087	–0.234	–0.429	0.328
26	Marsupial mouse, dusky	0.069	0.070	0.614	0.148	0.500	0.404	0.792	0.514	–3.17	–0.234	1.08	–0.825
27	Mink	0.284	–0.379	0.297	–0.078	–0.516	–0.208	0.511	–0.245	–0.650	0.439	–0.154	0.899
28	Mole	0.468	0.761	–1.10	–0.020	1.10	0.559	0.511	0.738	1.80	0.619	–1.31	0.943
29	Mole rat	1.43	0.462	–0.806	0.744	1.03	0.742	0.402	0.790	0.260	–0.374	–1.48	1.02
30	Monkey, green	0.010	0.520	–0.760	–0.327	–0.516	0.698	1.09	0.544	0.751	0.240	–1.82	1.53
31	Monkey, macaque	0.468	0.491	–1.73	–0.327	–0.838	0.404	0.611	0.082	0.461	1.55	–1.23	1.32
32	Mountain beaver	–0.607	–0.523	2.06	0.868	–1.45	–0.721	–0.327	–1.14	0.422	–0.374	–0.786	0.808
33	Mouse	1.73	0.576	0.907	1.60	1.75	2.76	–2.96	2.51	0.646	0.240	–1.25	–1.35
34	Mouse, spiny	0.309	–0.678	–0.391	–0.513	2.73	1.81	–0.782	2.07	1.18	0.703	0.912	–0.457
35	Opossum, didelphis	0.854	1.45	1.14	1.39	–0.240	0.404	0.013	0.044	1.24	1.50	–1.38	–2.06
36	Opossum, marmosa	–0.262	–0.474	–0.630	–0.913	–0.240	–0.046	0.013	–0.245	–0.101	–1.54	1.69	–1.85
37	Opossum, monodelphis	0.402	0.548	–0.247	0.107	0.217	–0.602	–1.47	–0.640	–0.253	0.619	1.27	–1.53
38	Possum, honey	–0.262	0.402	0.216	–0.078	0.668	0.653	0.511	0.630	0.915	1.14	0.790	–1.08
39	Possum, leadbeaters	–1.15	0.308	0.013	–0.513	0.316	–0.208	0.156	–0.074	–0.101	–0.104	1.05	–0.457
40	Possum, ring-tail	–0.423	0.432	0.013	–0.231	0.316	–0.125	–0.146	–0.115	–0.306	–0.374	0.751	–0.032
41	Possum, striped	–1.15	–0.789	–0.760	–1.52	–0.516	–1.16	–0.535	–1.07	0.302	0.342	1.04	0.133
42	Pouched mouse, jumping	–0.381	0.243	–0.213	–0.427	1.28	0.165	–0.782	0.392	1.01	0.935	1.14	–1.66
43	Prairie dog	–0.340	–0.162	0.073	–0.444	0.217	–0.125	0.013	–0.115	1.18	1.14	0.505	–0.093
44	Quoll	–0.558	–1.10	–0.547	–1.27	–1.23	0.509	–0.327	–0.201	0.499	1.14	1.28	–0.840
45	Rabbit	1.09	0.576	0.945	1.09	0.002	0.165	0.611	0.155	0.218	0.018	–0.284	–0.032
46	Raccoon	1.50	1.17	0.243	1.36	0.585	1.51	0.951	1.32	0.383	0.531	–1.22	0.086
47	Rat, albino	–1.60	1.11	0.216	0.009	1.03	–0.028	–0.327	0.260	–0.151	–0.234	0.744	–0.336
48	Rat, Norway	0.010	–0.573	–0.630	–0.792	1.16	0.865	–0.327	0.790	2.48	1.88	0.589	–1.49
49	Sheep	–0.053	–0.678	–0.017	–0.548	–0.838	0.228	0.402	–0.115	–0.151	0.018	–0.320	1.12
50	Squirrel	–0.119	–0.426	0.270	–0.327	0.112	–0.492	0.402	–0.201	1.21	1.38	0.547	–0.267
51	Tree shrew	–0.607	0.883	0.868	0.428	0.410	–1.55	0.156	–0.482	–0.151	0.240	–0.146	0.582
52	Wallaby, Little Rock	–1.50	0.631	0.375	–0.184	–1.71	–0.996	–1.08	–1.63	–1.05	–1.07	0.912	0.067
53	Wallaby, swamp	–1.40	–0.789	0.614	–0.621	–1.23	–0.492	–0.782	–1.07	–0.650	–0.870	0.816	0.337

In each column, mean is 0, standard deviation is 1, standard error of mean is 0.137.

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Table 4. A, B Animal rankings of each nucleus and two nuclear clusters

AVCN	PVCN	DCN	CN (AVCN + PVCN + DCN)	MTB	LSO
Bat, little brown	Bushbaby	Glider, feathertailed**	Glider, feathertailed*	Mouse, spiny**	Mouse**
Gopher, pocket	Kangaroo rat	Mountain beaver*	Gopher, pocket	Kangaroo rat*	Gropher, pocket
Mouse	Opossum, didelphis	Guinea pig	Bushbaby	Mouse	Mouse, spiny
Lemur, brown	Human	Chinchilla	Mouse	Bat, little brown	Lemur, brown
Raccoon	Gopher, pocket	Opossum, didelphis	Opossum, didelphis	Pouched mouse, jumping	Raccoon
Mole rat	Guinea pig	Bushbaby	Raccoon	Gopher, pocket	Fox, red
Kangaroo rat	Raccoon	Colocolo	Guinea pig	Rat, Norway	Loris, slow
Bushbaby	Rat, albino	Gopher, pocket	Kangaroo rat	Mole	Bat, little brown
Loris, slow	Tree shrew	Rabbit	Rabbit	Mole rat	Kangaroo rat
Rabbit	Chinchilla	Mouse	Bat, little brown	Rat, albino	Rat, Norway
Human	Loris, slow	Tree shrew	Mountain beaver	Capybara	Mole rat
Fox, red	Mole	Marsupial mouse, dusky	Human	Possum, honey	Monkey, green
Opossum, didelphis	Wallaby, Little Rock	Wallaby, swamp	Lemur, brown	Lemur, brown	Possum, honey
Armadillo	Lemur, brown	Armadillo	Mole rat	Raccoon	Bushbaby
Hedgehog	Mouse	Wallaby, Little Rock	Chinchilla	Colocolo	Mole
Mole	Rabbit	Bandicoot	Loris, slow	Loris, slow	Quoll
Monkey, macaque	Opossum, monodelphis	Llama	Tree shrew	Marsupial mouse, dusky	Marsupial mouse, dusky
Opossum, monodelphis	Monkey, green	Mink	Armadillo	Deer	Monkey, macaque
Deer	Cat	Squirrel	Colocolo	Fox, red	Opossum, didelphis
Mouse, spiny	Monkey, macaque	Raccoon	Fox, red	Tree shrew	Sheep
Mink	Mole rat	Possum, honey	Marsupial mouse, dusky	Possum, leadbeaters	Pouched mouse, jumping
Marsupial mouse, dusky	Possum, ring-tail	Rat, albino	Opossum, monodelphis	Possum, ring-tail	Rabbit
Capybara	Colocolo	Glider, sugar	Rat, albino	Bandicoot	Hedgehog
Monkey, green	Possum, honey	Prairie dog	**Average**	Opossum, monodelphis	Rat, albino
Rat, Norway	Bandicoot	Deer	Mole	Prairie dog	**Average**
Average	Hedgehog	Possum, leadbeaters	Mink	Beaver	Llama
Sheep	Possum, leadbeaters	Possum, ring-tail	Possum, honey	Squirrel	Opossum, marmosa
Llama	Glider, sugar	**Average**	Bandicoot	**Average**	Guinea pig
Beaver	Pouched mouse, jumping	Sheep	Wallaby, Little Rock	Rabbit	Human
Squirrel	Marsupial mouse, dusky	Capybara	Possum, ring-tail	Hedgehog	Possum, ring-tail
Opossum, marmosa	**Average**	Fox, red	Hedgehog	Llama	Prairie dog
Possum, honey	Armadillo	Pouched mouse, jumping	Monkey, green	Opossum, didelphis	Mink
Guinea pig	Prairie dog	Opossum, monodelphis	Monkey, macaque	Opossum, marmosa	Possum, leadbeaters
Prairie dog	Fox, red	Cat	Squirrel	Bushbaby	Bandicoot
Bandicoot	Glider, yellow-bellied	Mouse, spiny	Cat	Glider, feathertailed	Cat
Cat	Mink	Beaver	Glider, sugar	Glider, sugar	Colocolo
Pouched mouse, jumping	Squirrel	Human	Pouched mouse, jumping	Guinea pig	Chinchilla
Colocolo	Opossum, marmosa	Quoll	Deer	Mink	Squirrel
Possum, ring-tail	Mountain beaver	Glider, yellow-bellied	Prairie dog	Monkey, green	Wallaby, swamp
Quoll	Rat, Norway	Opossum, marmosa	Llama	Possum, striped	Glider, yellow-bellied
Chinchilla	Mouse, spiny	Rat, Norway	Possum, leadbeaters	Armadillo	Kangaroo, eastern grey
Mountain beaver	Sheep	Kangaroo, eastern grey	Mouse, spiny	Cat	Opossum, monodelphis
Tree shrew	Possum, striped	Bat, little brown	Sheep	Chinchilla	Armadillo
Dunnart, darling downs	Wallaby, swamp	Monkey, green	Wallaby, swamp	Monkey, macaque	Capybara
Glider, sugar	Dunnart, darling downs	Possum, striped	Rat, Norway	Sheep	Mountain beaver
Possum, leadbeaters	Kangaroo, eastern grey	Mole rat	Opossum, marmosa	Glider, yellow-bellied	Wallaby, Little Rock
Possum, striped	Bat, little brown	Mole	Capybara	Dunnart, darling downs	Dunnart, darling downs
Glider, yellow-bellied	Quoll	Kangaroo rat	Beaver	Quoll	Glider, feathertailed
Wallaby, swamp	Glider, feathertailed	Dunnart, darling downs	Quoll	Wallaby, swamp	Possum, striped
Wallaby, Little Rock	Llama	Glider, greater	Glider, yellow-bellied	Human	Glider, greater
Kangaroo, eastern grey	Deer	Lemur, brown	Possum, striped	Mountain beaver	Tree shrew
Rat, albino	Glider, greater*	Hedgehog	Kangaroo, eastern grey	Glider, greater	Beaver
Glider, feathertailed	Beaver*	Loris, slow	Dunnart, darling downs	Wallaby, Little Rock	Glider, sugar
Glider, greater*	Capybara**	Monkey, macaque	Glider, greater	Kangaroo, eastern grey	Deer*

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Table 4. B

MSO	SO (MTB + LSO + MSO)	VLL	DLL	IC	MG
Fox, red*	Mouse*	Rat, Norway*	Rat, Norway	Opossum, marmosa	Monkey, green
Kangaroo rat	Mouse, spiny*	Mole	Gopher, pocket	Dunnart, darling downs	Kangaroo, eastern grey
Lemur, brown	Kangaroo rat	Opossum, didelphis	Kangaroo rat	Quoll	Cat
Loris, slow	Fox, red	Squirrel	Monkey, macaque	Opossum, monodelphis	Monkey, macaque
Bushbaby	Gopher, pocket	Mouse, spiny	Opossum, didelphis	Pouched mouse, jumping	Bat, little brown
Bat, little brown	Lemur, brown	Prairie dog	Squirrel	Bandicoot	Capybara
Monkey, green	Bat, little brown	Pouched mouse, jumping	Possum, honey	Marsupial mouse, dusky	Sheep
Raccoon	Raccoon	Possum, honey	Prairie dog	Possum, leadbeaters	Deer
Cat	Loris, slow	Monkey, green	Quoll	Possum, striped	Glider, greater
Llama	Mole rat	Fox, red	Pouched mouse, jumping	Glider, yellow-bellied	Mole rat
Marsupial mouse, dusky	Rat, Norway	Lemur, brown	Mouse, spiny	Colocolo	Hedgehog
Human	Mole	Mouse	Fox, red	Mouse, spiny	Glider, sugar
Monkey, macaque	Possum, honey	Hedgehog	Mole	Wallaby, Little Rock	Mole
Rabbit	Bushbaby	Quoll	Opossum, monodelphis	Armadillo	Mink
Mink	Monkey, green	Bandicoot	Bushbaby	Wallaby, swamp	Human
Mole	Marsupial mouse, dusky	Monkey, macaque	Raccoon	Possum, honey	Mountain beaver
Possum, honey	Pouched mouse, jumping	Mountain beaver	Armadillo	Possum, ring-tail	Llama
Gopher, pocket	Rat, albino	Gopher, pocket	Human	Glider, greater	Beaver
Mole rat	Rabbit	Raccoon	Mink	Rat, albino	Tree shrew
Sheep	Llama	Possum, striped	Possum, striped	Beaver	Lemur, brown
Squirrel	Monkey, macaque	Guinea pig	Dunnart, darling downs	Rat, Norway	Guinea pig
Colocolo	Opossum, didelphis	Mole rat	Hedgehog	Squirrel	Wallaby, swamp
Guinea pig	**Average**	Kangaroo rat	Monkey, green	Prairie dog	Loris, slow
Possum, leadbeaters	Colocolo	Rabbit	Mouse	Llama	Glider, yellow-bellied
Tree shrew	Possum, leadbeaters	Loris, slow	Tree shrew	Glider, sugar	Fox, red
Chinchilla	Possum, ring-tail	Dunnart, darling downs	Glider, sugar	Deer	Possum, striped
Opossum, didelphis	Prairie dog	Bushbaby	Capybara	**Average**	Raccoon
Opossum, marmosa	Sheep	**Average**	Lemur, brown	Capybara	Wallaby, Little Rock
Prairie dog	Capybara	Glider, yellow-bellied	Rabbit	Kangaroo, eastern grey	**Average**
Average	Cat	Opossum, marmosa	Sheep	Tree shrew	Possum, ring-tail
Capybara	Quoll	Possum, leadbeaters	**Average**	Mink	Rabbit
Possum, ring-tail	Squirrel	Rat, albino	Possum, leadbeaters	Kangaroo rat	Chinchilla
Beaver	Guinea pig	Sheep	Loris, slow	Rabbit	Prairie dog
Deer	Mink	Tree shrew	Marsupial mouse, dusky	Sheep	Bushbaby
Mountain beaver	Opossum, marmosa	Chinchilla	Rat, albino	Hedgehog	Squirrel
Quoll	Bandicoot	Beaver	Bat, little brown	Loris, slow	Rat, albino
Rat, albino	Human	Opossum, monodelphis	Cat	Glider, feathertailed	Armadillo
Rat, Norway	Tree shrew	Glider, feathertailed	Mole rat	Fox, red	Mouse, spiny
Bandicoot	Hedgehog	Possum, ring-tail	Mountain beaver	Guinea pig	Possum, leadbeaters
Glider, yellow-bellied	Chinchilla	Llama	Possum, ring-tail	Cat	Colocolo
Glider, sugar	Opossum, monodelphis	Colocolo	Glider, yellow-bellied	Mountain beaver	Dunnart, darling downs
Possum, striped	Deer	Kangaroo, eastern grey	Kangaroo, eastern grey	Human	Marsupial mouse, dusky
Armadillo	Beaver	Human	Bandicoot	Bushbaby	Quoll
Dunnart, darling downs	Armadillo	Mink	Deer	Gopher, pocket	Bandicoot
Mouse, spiny	Glider, yellow-bellied	Wallaby, swamp	Glider, greater	Raccoon	Possum, honey
Pouched mouse, jumping	Possum, striped	Capybara	Wallaby, swamp	Monkey, macaque	Kangaroo rat
Wallaby, swamp	Wallaby, swamp	Glider, greater	Wallaby, Little Rock	Mouse	Mouse
Glider, feathertailed	Glider, feathertailed	Cat	Beaver	Lemur, brown	Glider, feathertailed
Kangaroo, eastern grey	Mountain beaver	Bat, little brown	Glider, feathertailed	Mole	Rat, Norway
Wallaby, Little Rock	Glider, sugar	Wallaby, Little Rock	Guinea pig	Chinchilla	Opossum, monodelphis
Glider, greater	Dunnart, darling downs	Glider, sugar	Colocolo	Opossum, didelphis	Pouched mouse, jumping
Opossum, monodelphis	Kangaroo, eastern grey	Armadillo	Chinchilla	Mole rat	Opossum, marmosa
Hedgehog**	Wallaby, Little Rock	Deer**	Opossum, marmosa	Monkey, green	Gopher, pocket
Mouse**	Glider, greater	Marsupial mouse, dusky**	Llama**	Bat, little brown**	Opossum, didelphis*

CN = AVCN + PVCN + DCN and SO = MTB + LSO + MSO based on entries in table 3: largest at top, smallest at bottom. Animal ranks less than 6 apart are not reliably different.

* = $p < 0.05$; ** = $p < 0.01$.

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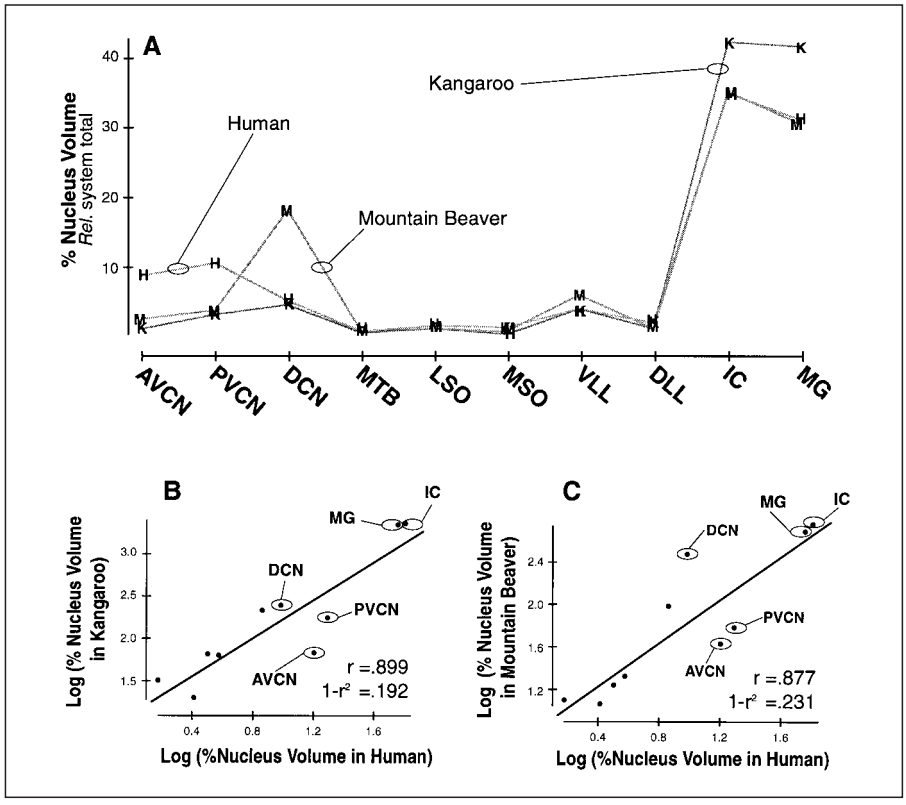


Fig. 5. 'Form' of the brainstem auditory system defined as the collective 'profile' of volume x nucleus. **A** Three profiles of form of ascending auditory system: human (H), mountain beaver (M) and kangaroo (K). Each line interconnects the percentage sizes for the 10 nuclei, one line for each of the three animals. **B** and **C** Scatterplots of relative sizes of 10 nuclei in human vs. kangaroo (**B**) and human vs. mountain beaver (**C**). In the scatterplots, each point represents one of the 10 auditory nuclei. Correlation coefficients, r, derived from long-normalized data in **B** and **C** can be used as an estimate of 'similarity in overall form' relatively unbiased by large differences in size of the individual nuclei (e.g., see IC or MG vs. DLL or MSO in **A**). The circled points in **B** and **C** indicate the five largest individual nuclei in humans.

geological past. That is, a special ecological demand in a remote geological time can leave a stamp on a nucleus (or on the entire system) that has not necessarily eroded over the long sequence of re-adaptations to successively more recent ecologies. Alternatively, such structural differences may have accumulated slowly but at nearly equal rates over the time since the original divergence of the taxa. That these possibilities are indeed present in the data is indicated by noting that several of the nuclei are reliably different among the two subclasses represented (i.e., marsupials vs. placentals) but not among the taxa at lesser levels. For example, medial geniculate (MG) is reliably different between the marsupials and placentals in the sample ($F_{51}^{1} = 9.0147$; $p < 0.004$, in the present data) [Glendenning and Masterton, 1994; Masterton and Glendenning, 1994]. However, MG is *not* reliably different either among successive orders, families or genera within the two subclasses ($p > 0.05$). Marsupials and placentals diverged sometime before the Paleocene and possibly as long ago as the Jurassic. Therefore, these differences in the size of MG may have occurred near the time of divergence, or they may have accumulated due to differing selective pressures over the subsequent 70 to 200 million years, or both. In either case, the matching of presently deviant nuclei with presently deviant ecologies must await the statistical removal from the data of the effects of long segments of evolutionary history. Only differences at lower taxonomic levels after differences among the higher taxa are removed, hence differences gained in more recent times, might be expected to correspond to present ecological differences. This seemingly formidable task can be partly realized by so-called hierarchical analysis of variance among successively lower taxonomic levels (i.e., orders within subclasses, families within orders, genera within families, etc.). These taxonomic or historical analyses of the present data are the subject of a future report.

Form of the Auditory System

To illustrate our usage of the term 'form' in the present context, the percentages of the 10 nuclei for human, together with those for kangaroos and mountain beavers for contrast, are shown in figure 5. It should be noted that each of the three profiles in figure 5 depicts the overall 'form' of that animal's auditory system.

It can be seen in figure 5A that the three animals have similar though not identical profiles or 'forms'. All three

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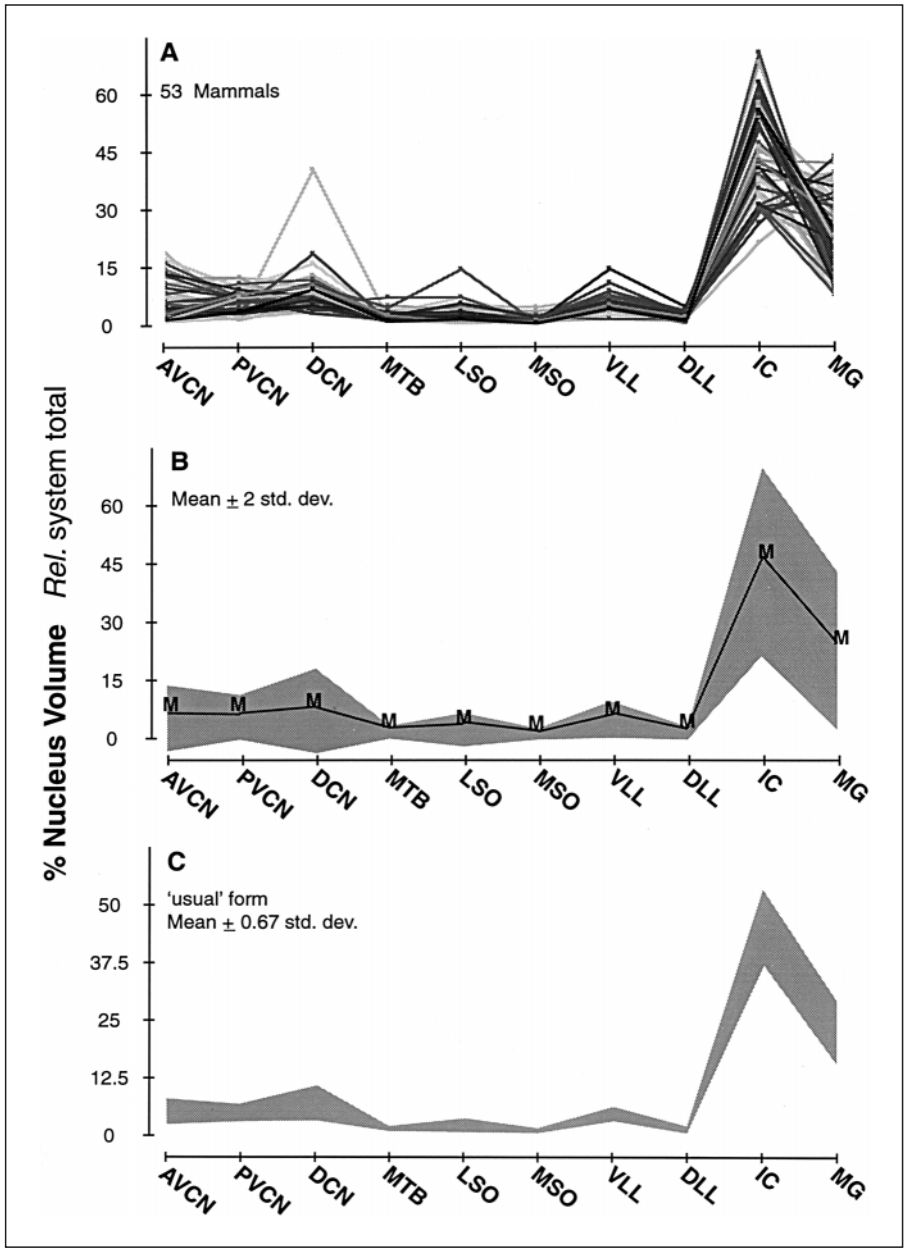


Fig. 6. Variation in form of brainstem auditory system among 53 viviparous mammals. **A** Line plots of sizes of the 10 nuclei in the 53 animals. Each line represents one animal; each point interconnected by one line represents the relative size of the nucleus in that animal. **B** Statistically derived mean (M) and dispersion (mean ± 2 std. dev.) of each nucleus. Therefore, the gray area contains about 95% of the animals in the sample. **C** 'Usual' form of ascending auditory system. The gray area contains about 50% of the animals in sample.

have a large IC and MG. However, humans ('H') deviate noticeably from kangaroos ('K') in the size of two of their cochlear nuclei (AVCN, PVCN) and again in the size of their IC and MG. In contrast, mountain beavers ('M') parallel humans, except for their cochlear nuclei and VLL. These differences are more easily seen in the two scatterplots (fig. 5B, C).

Figure 6A shows the same sort of graph as in figure 5 but this time for all 53 animals. That is, there are 53 lines, one line for each animal. Each line connects 10 points, one point for each nucleus. Loosely defined by concentrations of near-coincident lines, figure 6A shows first that there is indeed a 'usual' form of the ascending auditory system and second that there are a few obvious deviations from that form.

Figure 6B is a statistical summary of figure 6A. It consists of a line graph of nucleus means and a shaded area surrounding it, both statistically derived from table 2. The 'M's represent the average or mean percentage of each nucleus for the 53 animals. The shaded area around them rep-

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resents the mean ±2 standard deviations. Therefore, the shaded area contains about 95% of the animals in the sample (i.e., about 50/53). This wide area can be used as a template for identifying a particular animal as having a deviant nucleus (see table 3). However, for the present purpose of estimating the usual form of the auditory system in the present sample of animals, we have calculated a narrower range – the mean ±0.67 std. dev. – which contains about 50% of the animals in the sample for each nucleus. This narrower profile is shown in figure 6C. It will be referred to as the 'usual form' in the remainder of this report.

Figure 6B shows that on average the relative sizes of the 10 nuclei are well ordered, with IC the largest nucleus by far and MSO the smallest. The relative volume of IC is almost half (47.6%) that of the total system, usually twice that of MG which is the second largest nucleus, and 2.5 times the size of DCN, the third largest nucleus. The three cochlear nuclei together (AVCN + PVCN + DCN) constitute almost 20% and MG alone is 23.5% of the total system. The cochlear nuclei, IC, and MG together make up over 90% of the total system.

Deviations from the 'Common Form'

Given the statistically derived 'usual form' (fig. 6C), animals most deviant and least deviant can be identified using a statistic that measures their degree of similarity to the usual form. One way of establishing each animal's degree of similarity to, or deviation from, the common form while maintaining unit-less numbers is by calculating the least-squares correlation coefficient between the average form and the form of each animal alone. Calculated from the log normalized data in table 3 to give each nucleus more equal weight, the resulting correlation, 'r', is a relatively unbiased estimate of the degree of similarity of each animal's form to the usual form and therefore can be used as a 'similarity index' with values ranging from 1 (perfect identity with the usual form) through 0 (no similarity) to –1 (inverse identity).

The 'similarity index' of each animal's form with the usual form is listed in table 5. The animals are listed both alphabetically (i.e., in the same order as they appear in previous tables) and also in terms of decreasing similarity to (or increasing deviation from) the usual form. The 53 'similarity indices' range from a low of 0.686 to a high of 0.990 with an average similarity of 0.923. This high average value means, once more, that the form of the auditory system is quite similar for most animals in the sample (cf. fig. 6C). Biologically, this high degree of similarity among mammals suggests that the overall form of the auditory system has been relatively stable over geological time. And insofar as long periods of geological time allow wide variations in ecology, the auditory system seems notably resistant to great change, at least among the mammals in the present sample [cf. Bang and Cobb, 1968]. We return to this point below.

The animals with the least deviation from the average form are at the top of the right list in table 5. They are ringtailed possums, bandicoots, and yellow-bellied gliders – all marsupials – with indices of 0.990, 0.980, and 0.977 and with deviations ($1-r^2$) of only 2, 4, and 4.6%, respectively. Profiles of these 'least deviant' forms are shown in figure 7A.

Animals with Deviant Forms

The common form shown in figure 6C also allows the identification of the most deviant forms. Specifically, the form of an animal's auditory system is the most 'deviant' if its particular form is least similar to the average form. The animals with the lowest 'similarity indices' with the average form are shown at the bottom of the right list in table 5. The three most deviant forms in the sample are mouse ($r = 0.686$), bats ($r = 0.782$), and kangaroo rats ($r = 0.811$) – all placentals – with deviations of 47, 38.9, and 34.3%, respectively. Profiles of these most deviant animals are shown in figure 7B.

Perhaps surprisingly, the mountain beaver, with its large DCN (fig. 5), is *not* among these 'most deviant' forms: It is 32nd most deviant. The reason is that the DCN of the mountain beaver is its only notable deviation from the usual form [see Merzenich et al., 1973]. Its DCN, though 'deviant' itself, is not deviant enough to override the similarity of the remaining nuclei in its auditory system (see table 2). For this same reason the feathertailed glider, whose DCN dwarfs even the mountain beaver's, has only the 8th most deviant form [see table 2; also see Aitkin and Nelson, 1989]. Therefore, although a 'deviant form', as defined here, depends on the presence of 'deviant nuclei', the two kinds of deviation are not identical. We return to this question below.

Form of Two Auditory Subsystems

To this point we have described the average size and the 'usual form' of the 10 largest nuclei in the subcortical auditory system (fig. 6A, B, C) and the animals whose forms are least deviant (fig. 7A) and most deviant (fig. 7B). This same type of analysis can be applied to subparts of the auditory system as well. In the next section, we describe the variations among the three sound-localization nuclei within the superior olivary complex. Here we describe the variations among the three nuclei within the cochlear nucleus: namely, AVCN, PVCN, and DCN.

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Table 5. Animal rankings based on similarity in form of each animal's ascending auditory system with average form of system

Alphabetical order	r	Descending order	r	
Armadillo	0.957	Possum, ring-tail	0.990	least deviant
Bandicoot	0.980	Bandicoot	0.980	
Bat, little brown	0.782	Glider, yellow-bellied	0.977	
Beaver	0.945	Mink	0.976	
Bushbaby	0.918	Prairie dog	0.976	
Capybara	0.916	Wallaby, swamp	0.976	
Cat	0.949	Sheep	0.972	
Colocolo	0.951	Dunnart, darling downs	0.971	
Chinchilla	0.944	Wallaby, Little Rock	0.969	
Deer	0.911	Possum, striped	0.968	
Dunnart, darling downs	0.971	Rabbit	0.968	
Fox, red	0.883	Kangaroo, eastern grey	0.967	
Glider, yellow-bellied	0.977	Possum, leadbeaters	0.966	
Glider, feathertailed	0.865	Possum, honey	0.965	
Glider, greater	0.962	Glider, greater	0.962	
Glider, sugar	0.954	Squirrel	0.961	
Gopher, pocket	0.813	Armadillo	0.957	
Guinea pig	0.945	Glider, sugar	0.954	
Hedgehog	0.824	Colocolo	0.951	
Human	0.925	Opossum, marmosa	0.951	
Kangaroo rat	0.811	Cat	0.949	
Kangaroo, eastern grey	0.967	Mountain beaver	0.946	
Lemur, brown	0.847	Beaver	0.945	
Llama	0.908	Guinea pig	0.945	
Loris, slow	0.878	Chinchilla	0.944	
Marsupial mouse, dusky	0.892	Tree shrew	0.943	
Mink	0.976	Mole	0.942	
Mole	0.942	Opossum, monodelphis	0.940	
Mole rat	0.919	Quoll	0.940	
Monkey, green	0.922	Rat, albino	0.928	
Monkey, macaque	0.903	Raccoon	0.927	
Mountain beaver	0.946	Human	0.925	
Mouse	0.686	Pouched mouse, jumping	0.924	
Mouse, spiny	0.852	**Average similarity**	**0.923**	
Opossum, didelphis	0.876	Monkey, green	0.922	
Opossum, marmosa	0.951	Mole rat	0.919	
Opossum, monodelphis	0.940	Bushbaby	0.918	
Possum, honey	0.965	Capybara	0.916	
Possum, leadbeaters	0.966	Deer	0.911	
Possum, ring-tail	0.990	Llama	0.908	
Possum, striped	0.968	Monkey, macaque	0.903	
Pouched mouse, jumping	0.924	Marsupial mouse, dusky	0.892	
Prairie dog	0.976	Fox, red	0.883	
Quoll	0.940	Loris, slow	0.878	
Rabbit	0.968	Opossum, didelphis	0.876	
Raccoon	0.927	Rat, Norway	0.867	
Rat, albino	0.928	Glider, feathertailed	0.865	
Rat, Norway	0.867	Mouse, spiny	0.852	
Sheep	0.972	Lemur, brown	0.847	
Squirrel	0.961	Hedgehog	0.824	
Tree shrew	0.943	Gopher, pocket	0.813	
Wallaby, Little Rock	0.969	Kangaroo rat	0.811	
Wallaby, swamp	0.976	Bat, little brown	0.782	
		Mouse	0.686	most deviant

Ranks less than 10 apart are not reliably different.

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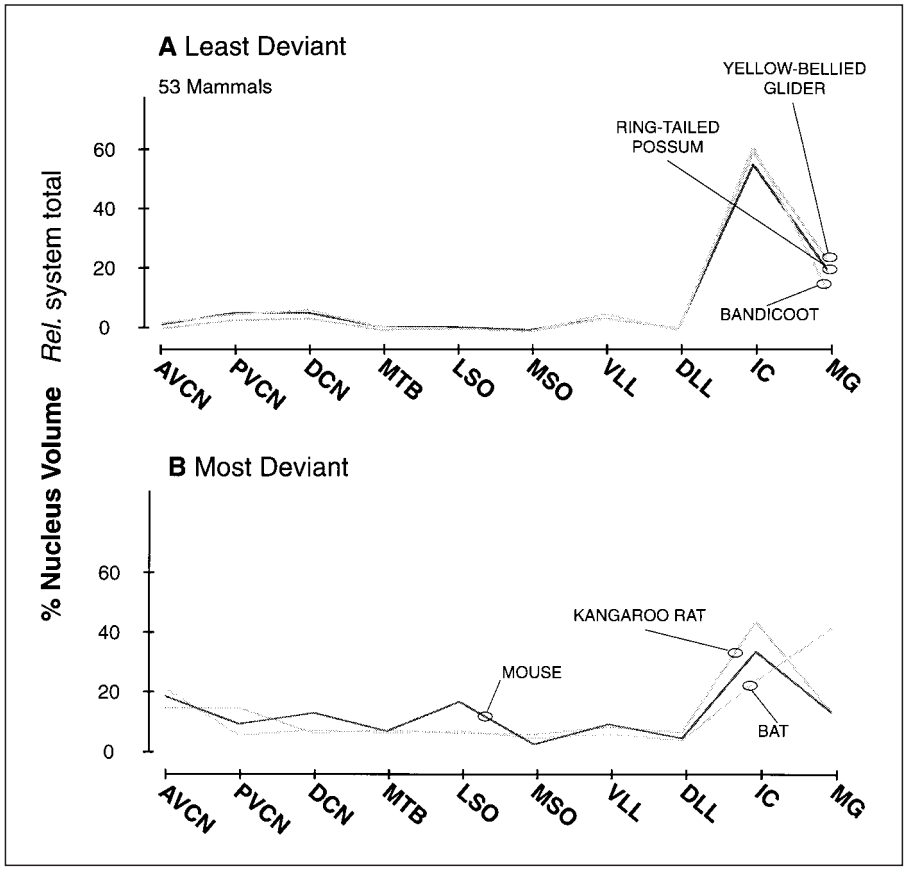


Fig. 7. Form of auditory system in 6 peculiar mammals relative to usual form. **A** Three animals whose ascending auditory system is least deviant (or most similar) to the usual form. **B** Three animals whose system is most deviant (or least similar) to the usual form. See figure 6C for definition for 'usual' form.

Form of the Cochlear Nucleus. Figure 8 shows the distribution of the size of the cochlear nucleus relative to the size of the entire auditory system (fig. 8A). Figure 8 also shows the distributions in size of AVCN, PVCN, and DCN, first relative to the size of the total system (fig. 8B–D) and then relative to the size of the entire cochlear nucleus itself (fig. 8E–G). It should be noted that these data are standardized by these procedures (as in table 2), but they are *not* normalized (as in table 3). Therefore, the distributions remain somewhat skewed.

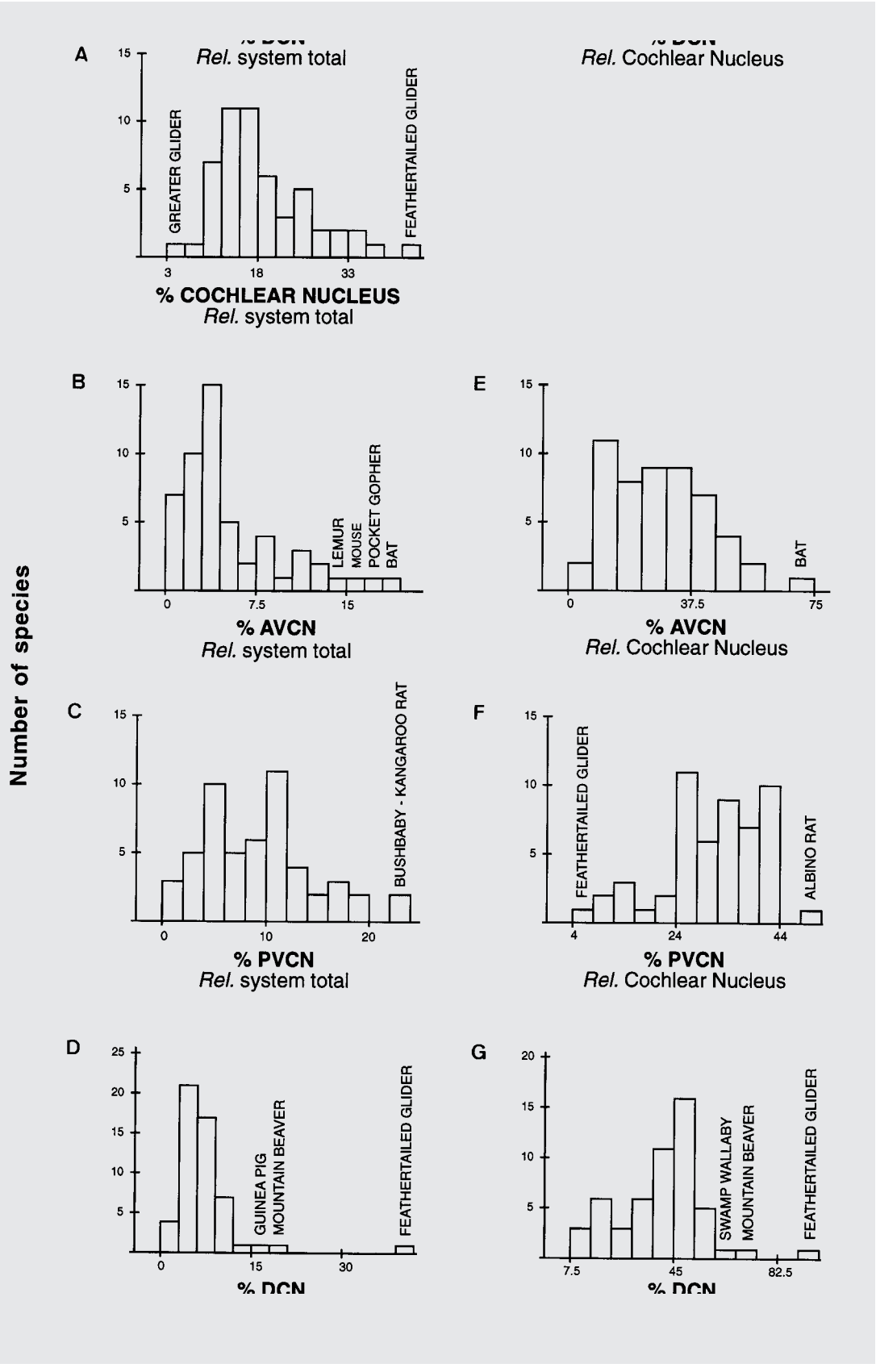
The variation in form of the cochlear nucleus can be visualized in figure 9. Figure 9A, like figure 6A, has 53 lines, one for each animal, and is simply a magnification of the first three points in figure 6A – that is, the size of the three nuclei relative to the size of the entire system. However, figure 9B is *not* a simple magnification of earlier profiles because in this graph, the divisor of each of the three nuclear volumes is the sum of the three volumes – the volume of the entire cochlear nucleus itself.

Despite the variation seen in figure 9B, a 'usual form' for the relative sizes of the three cochlear nuclei is evident. This form is shown in figure 9C – the gray area is demarcated by μ ±0.67 std. dev. once more – the area containing 50% of the animals in the sample. Although all three nuclei are nearly equal on average, DCN is slightly larger (38%), while PVCN (30%) and AVCN (32%) are slightly smaller.

Correlations of the form of each animal's cochlear nucleus with the usual form to determine their relative 'similarity' are shown in table 6. The greater glider, prairie dog and swamp wallaby – two marsupials, one placental – are the animals whose cochlear nuclei are the 'least deviant' based on the form of their cochlear nuclei alone. In contrast, lemur, raccoon and loris, each a placental, are the

Fig. 8. Distribution in sizes of the total Cochlear Nucleus (AVCN + PVCN + DCN) and of AVCN, PVCN and DCN alone relative to system total (left column) and relative to cochlear nucleus total (right column). Although standardized, the data in these histograms are not normalized. Therefore, the actual non-normal skew of the data is still visible.

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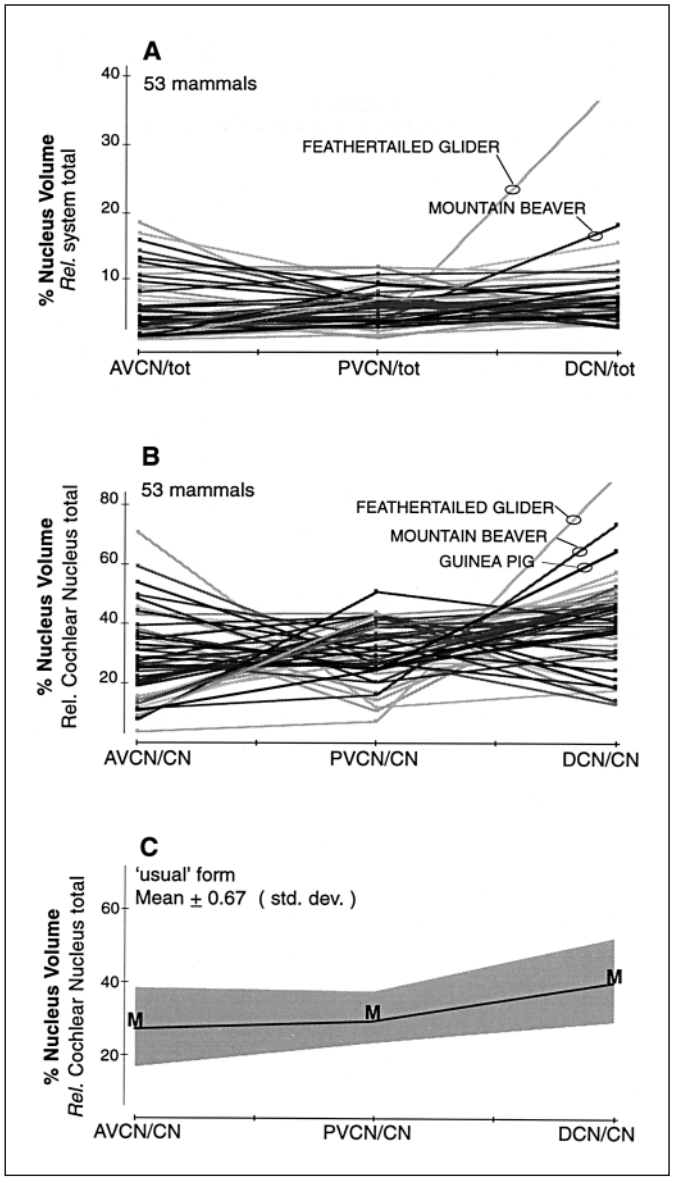


Fig. 9. Line plots of the three cochlear nuclei relative to auditory system total (**A**) and relative to cochlear nucleus total (**B**). **C** Statistically derived mean and area containing 50% of the sample shown in **B**.

animals with cochlear nuclei least similar to the average form and hence, have the 'most deviant' forms.

Form of the Superior Olives. The superior olivary complex (SO) of some mammals consists of 11 separable nuclei [cf. Morest, 1968, 1973; Warr, 1975, 1982; Moore, 1987; Spangler et al., 1987]. Eight of these nuclei originate olivocochlear and olivocochlear-nucleus fibers and are part of the descending auditory system [Warr, 1975, 1978, 1982; Spangler et al., 1987; Spangler and Warr, 1991]. The three remaining nuclei (MTB, LSO, MSO) are either all (MSO) or partly (MTB, LSO) constituents of the ascending system. Indeed, it is the ascending components of these three nuclei that are known to provide binaural analyses for sound localization [e.g., see Stotler, 1953; Irvine, 1986; Masterton, 1992].

Figure 10 shows the distribution of the size of the superior olives (i.e., MTB + LSO + MSO) relative to the size of the entire subcortical system (fig. 10A) and then the distribution of each of the three nuclei relative to the system total (fig. 10B–D), and, finally, relative to their own sum (fig. 10E–G).

The form of these nuclei relative to the entire auditory system is shown in figure 11A. Once more, figure 11A is a simple magnification of the 4th, 5th and 6th points in figure 6A. Figure 11B shows the same nuclei relative to their own sum (SO = MTB + LSO + MSO), while figure 11C shows the statistically derived 'usual form' – the mean ± 0.67 std. dev. for each nucleus. It can be seen in figure 11C that LSO averages the largest and MSO the smallest of the three nuclei.

Deriving measures of 'similarity' using correlation coefficients applied to the normalized data once more (table 7) shows that the animals whose superior olives deviate least from the usual form are mole rats, bats, hedgehogs, and Norway rats – all placentals – with similarity indices of 1.000 or 0.999. The most deviant animals represented are cats, foxes, and llamas – all placentals – with similarity indices of –0.710, –0.348, and –0.232 – the negatives indicating that the order of relative sizes of the three nuclei are the inverse of the usual form, with MSO, instead of LSO, being the largest of the three SOC nuclei.

General Discussion

Given the statistical derivations of the usual forms and sizes of auditory nuclei and the identification of the deviant animals in the sample, several questions as to the applicability of these conclusions arise. Perhaps the first of these is whether or not the present conclusions can be generalized to the entire class Mammalia. The other questions pertain to further analyses of the data (taxonomic, evolutionary, ecological, etc.) that might prove to make the present data more useful. These questions are discussed in order before the breadth of the present conclusions is discussed.

Representativeness of Sample

Although the sample size of 53 seems sufficiently large to gain some measure of confidence, its usefulness in estimating the characteristics of the auditory system in the pop-

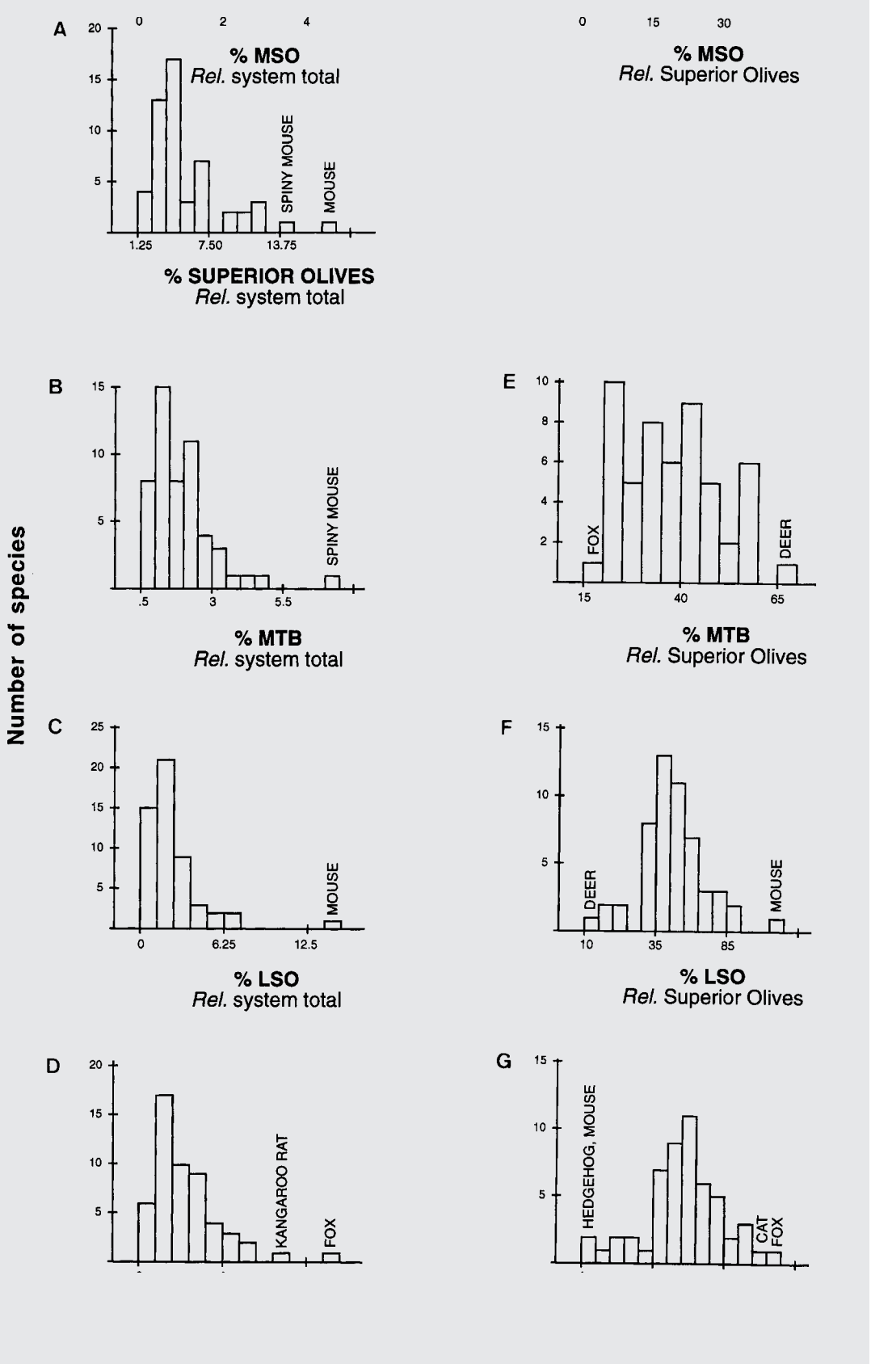
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Table 6. Animal rankings based on similarity in form of each animal's cochlear nuclei with average form of cochlear nuclei

Alphabetical order	r	Descending order	r	
Armadillo	0.133	Glider, greater	0.999	least deviant
Bandicoot	0.946	Prairie dog	0.999	
Bat, little brown	–0.685	Wallaby, swamp	0.999	
Beaver	0.310	Opossum, marmosa	0.998	
Bushbaby	0.228	Colocolo	0.986	
Capybara	0.257	Marsupial mouse, dusky	0.979	
Cat	0.650	Glider, feathertailed	0.978	
Colocolo	0.986	Mountain beaver	0.964	
Chinchilla	0.963	Chinchilla	0.963	
Deer	0.223	Guinea pig	0.963	
Dunnart, darling downs	0.894	Possum, striped	0.956	
Fox, red	–0.512	Bandicoot	0.946	
Glider, yellow-bellied	0.878	Kangaroo, eastern grey	0.945	
Glider, feathertailed	0.978	Quoll	0.927	
Glider, greater	0.999	Squirrel	0.927	
Glider, sugar	0.896	Possum, honey	0.919	
Gopher, pocket	–0.761	Opossum, didelphis	0.905	
Guinea pig	0.963	Tree shrew	0.904	
Hedgehog	–0.956	Glider, sugar	0.896	
Human	–0.750	Dunnart, darling downs	0.894	
Kangaroo rat	–0.898	Glider, yellow-bellied	0.878	
Kangaroo, eastern grey	0.945	Possum, leadbeaters	0.865	
Lemur, brown	–0.999	Wallaby, Little Rock	0.855	
Llama	0.707	Possum, ring-tail	0.841	
Loris, slow	–0.982	Pouched mouse, jumping	0.839	
Marsupial mouse, dusky	0.979	Sheep	0.777	
Mink	0.655	Rat, albino	0.767	
Mole	–0.717	Llama	0.707	
Mole rat	–0.973	Mink	0.655	
Monkey, green	–0.014	Cat	0.650	
Monkey, macaque	–0.861	Rat, Norway	0.534	
Mountain beaver	0.964	**Average similarity**	**0.356**	
Mouse	–0.459	Beaver	0.310	
Mouse, spiny	0.074	Capybara	0.257	
Opossum, didelphis	0.905	Bushbaby	0.228	
Opossum, marmosa	0.998	Deer	0.223	
Opossum, monodelphis	–0.074	Rabbit	0.221	
Possum, honey	0.919	Armadillo	0.133	
Possum, leadbeaters	0.865	Mouse, spiny	0.074	
Possum, ring-tail	0.841	Monkey, green	–0.014	
Possum, striped	0.956	Opossum, monodelphis	–0.074	
Pouched mouse, jumping	0.839	Mouse	–0.459	
Prairie dog	0.999	Fox, red	–0.512	
Quoll	0.927	Bat, little brown	–0.685	
Rabbit	0.221	Mole	–0.717	
Raccoon	–0.982	Human	–0.750	
Rat, albino	0.767	Gopher, pocket	–0.761	
Rat, Norway	0.534	Monkey, macaque	–0.861	
Sheep	0.777	Kangaroo rat	–0.898	
Squirrel	0.927	Hedgehog	–0.956	
Tree shrew	0.904	Mole rat	–0.973	
Wallaby, Little Rock	0.855	Loris, slow	–0.982	
Wallaby, swamp	0.999	Raccoon	–0.982	
		Lemur, brown	–0.999	most deviant

Ranks less than 12 apart are not reliably different.

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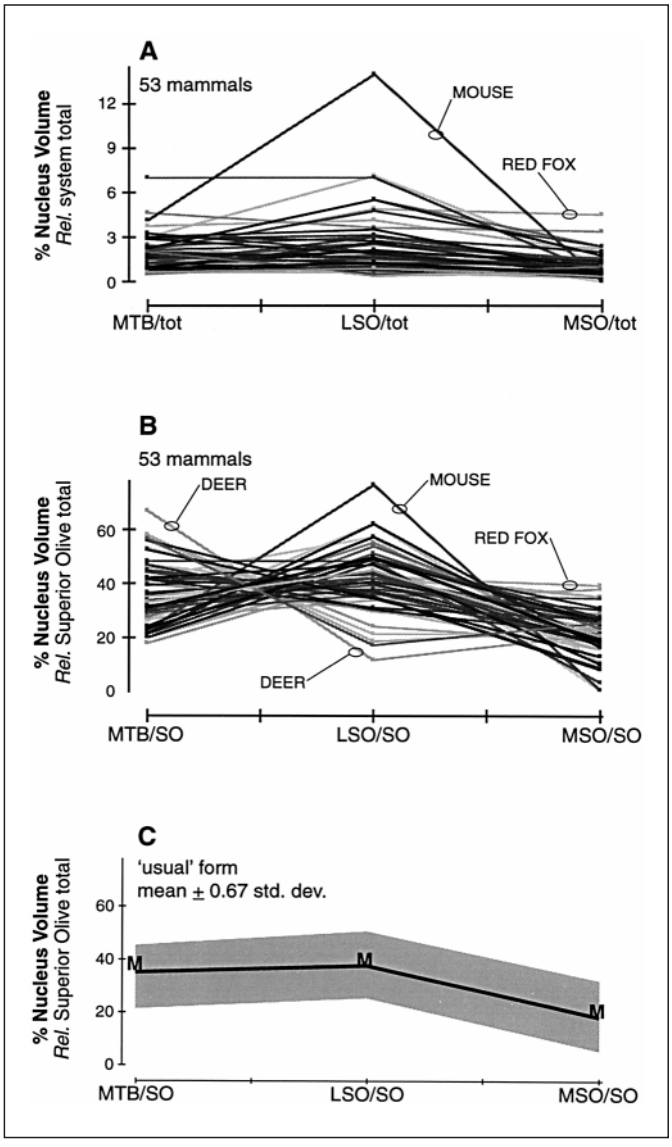


Fig. 11. Line plots of three superior olivary nuclei relative to auditory system total (**A**) and relative to superior olive total (**B**) (see captions to fig. 6 and 9). **C** Statistically derived mean and area containing 50% of the sample (from **B**).

Fig. 10. Standardized, non-normalized distributions in sizes of three superior olivary nuclei (MTB + LSO + MSO) and of MTB, LSO, and MSO alone relative to the system total (left column) and relative to superior olive total (right column). Peculiarity of Mouse in **A** reappears as a peculiarity of LSO in **C** and **F**.

ulation of viviparous mammals, marsupials and placentals together, depends on its randomness, not just its size.

Strictly speaking, any statistical inference from sample to population to which a precise probability statement can be attached usually requires a random sample. The problem of biological morphometry, however, is that a truly random sample of a large taxon, such as the class Mammalia, is simply not possible to achieve: a truly random sample will call for species that are not available due to their rarity, geographic isolation, endangered status, etc. Therefore, while we examined 53 diverse species, they do not constitute a random sample. Given the practical impossibility of true randomness, the next less strict question becomes: is the sample 'representative'? This question can be answered by comparing the distribution of the animals in the sample with the actual world-wide distribution of mammals.

Table 8 shows the distribution of viviparous mammals in terms of genera per order [genera being the lowest taxa which are usually unargued among taxonomists; Simpson, 1945]. Table 8 (Col. 1) shows the actual number of genera per order for the six orders of marsupials and for the 14 largest orders of placentals [cf. Nowak, 1991]. It also shows the number of genera per order in the present sample (Col. 4). Column 7 in the table shows the deviation of the sample percentage from the actual percentage of mammals, while column 8 shows the deviation within the two subclasses.

The table (Col. 7 = Col. 5 – Col. 2) shows that the present sample is not a representative sample of all mammals. To begin with, there are far too many marsupials in the sample (37.7% in Col. 5 vs. 7.1% in Col. 2). This overweighting of marsupials is no accident. It was done intentionally because one of the goals of the research program is to determine pervasive taxonomic differences among the animals. This vast over-representation of marsupials in the present sample will be exploited for that purpose in a further report.

Within the marsupials, however, the sample proves to be a fairly close representation of the actual distribution of marsupial genera per order. Comparing the percentages in columns 3 and 6 (Col. 8 = Col. 6 – Col. 3) in the table, the sample proves to be within 10% for each of the six orders of marsupials. Therefore, the present sample of mammals is seriously over-weighted with marsupials, but the sample of marsupials itself is reasonably representative (though still not random).

Turning to the placentals, there are 33 genera in the sample representing the actual 1,021 genera of placentals in the 14 largest orders. Column 7 (Col. 5 – Col. 2), showing the difference between the percentages in the sample and the worldwide percentages, indicates three notable deviations.

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Table 7. Animal rankings based on similarity in form of each animal's superior olives with average form of superior olives

Alphabetical order	r	Descending order	r	
Armadillo	0.989	Mole rat	1.000	least deviant
Bandicoot	0.956	Bat, little brown	0.999	
Bat, little brown	0.999	Hedgehog	0.999	
Beaver	0.210	Rat, Norway	0.999	
Bushbaby	–0.173	Opossum, marmosa	0.997	
Capybara	0.622	Mouse	0.995	
Cat	–0.710	Mouse, spiny	0.995	
Colocolo	0.722	Possum, honey	0.990	
Chinchilla	0.953	Armadillo	0.989	
Deer	0.062	Dunnart, darling downs	0.988	
Dunnart, darling downs	0.988	Glider, greater	0.988	
Fox, red	–0.348	Mole	0.988	
Glider, yellow-bellied	0.987	Glider, yellow-bellied	0.987	
Glider, feathertailed	0.847	Prairie dog	0.973	
Glider, greater	0.988	Marsupial mouse, dusky	0.970	
Glider, sugar	0.389	Bandicoot	0.956	
Gopher, pocket	0.918	Possum, ring-tail	0.954	
Guinea pig	0.953	Chinchilla	0.953	
Hedgehog	0.999	Guinea pig	0.953	
Human	–0.142	Wallaby, Little Rock	0.951	
Kangaroo rat	0.568	Pouched mouse, jumping	0.945	
Kangaroo, eastern grey	0.873	Opossum, monodelphis	0.944	
Lemur, brown	0.541	Wallaby, swamp	0.930	
Llama	–0.232	Rat, albino	0.921	
Loris, slow	0.495	Gopher, pocket	0.918	
Marsupial mouse, dusky	0.970	Opossum, didelphis	0.917	
Mink	0.667	Possum, leadbeaters	0.904	
Mole	0.988	Rabbit	0.880	
Mole rat	1.000	Kangaroo, eastern grey	0.873	
Monkey, green	0.091	Glider, feathertailed	0.847	
Monkey, macaque	0.346	Possum, striped	0.776	
Mountain beaver	0.715	Raccoon	0.758	
Mouse	0.995	Quoll	0.743	
Mouse, spiny	0.995	Colocolo	0.722	
Opossum, didelphis	0.917	Mountain beaver	0.715	
Opossum, marmosa	0.997	**Average similarity**	**0.668**	
Opossum, monodelphis	0.944	Mink	0.667	
Possum, honey	0.990	Squirrel	0.662	
Possum, leadbeaters	0.904	Capybara	0.622	
Possum, ring-tail	0.954	Kangaroo rat	0.568	
Possum, striped	0.776	Lemur, brown	0.541	
Pouched mouse, jumping	0.945	Loris, slow	0.495	
Prairie dog	0.973	Sheep	0.480	
Quoll	0.743	Glider, sugar	0.389	
Rabbit	0.880	Monkey, macaque	0.346	
Raccoon	0.758	Beaver	0.210	
Rat, albino	0.921	Monkey, green	0.091	
Rat, Norway	0.999	Tree shrew	0.083	
Sheep	0.480	Deer	0.062	
Squirrel	0.662	Human	–0.142	
Tree shrew	0.083	Bushbaby	–0.173	
Wallaby, Little Rock	0.951	Llama	–0.232	
Wallaby, swamp	0.930	Fox, red	–0.348	
		Cat	–0.710	most deviant

Ranks less than 12 apart are not reliably different.

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Table 8. Actual and sample distributions of major mammalian orders (in genera per order)

Order	Actual No.	% of mammals	% of marsupials	Sample No.	% of mammals	% of marsupials	% difference among mammals (sample–actual) col 5–col 2	% difference within subclasses col 6–col 3
Subclass marsupial								
Didelphiformes	12	15.4	15.4	4	7.5	20.0	+6.4*	+4.6
Paucituberculata	2	2.6	2.6	0	0	0	–0.2	–2.6
Dasyurida	19	1.7	24.4	4	7.5	20.0	+5.8*	–4.4
Peramelina	8	0.7	10.3	1	1.9	5.0	+1.2	–5.3
Notoryctemorphia	1	0.1	1.3	0	0	0	–0.1	–1.3
Diprotodonta	36	3.3	46.2	11	20.8	55.0	+17.5*	+8.8
Total marsupials	78	7.1		20	37.7			
			% of placentals			% of placentals		
Subclass placental								
Insectivora	65	5.9	6.4	2	3.8	6.1	–2.1	–0.3
Dermoptera	1	0.1	0.1	0	0	0	–0.1	–0.1
Chiroptera	186	16.9	18.2	1	1.9	3.0	–15.0*	–15.2
Scandentia	5	0.5	0.5	1	1.9	3.0	+1.4	+2.5
Primates	71	6.5	7.0	6	11.3	18.2	+4.8	+11.2
Xenarthra	13	1.7	1.3	1	1.9	3.0	+0.2	+1.7
Lagomorpha	13	1.2	1.3	1	1.9	3.0	+0.7	+1.7
Rodentia	426	38.8	41.7	14	26.4	42.4	–12.4*	+0.7
Cetacea	41	3.7	4.0	0	0	0	–3.7	–4.0
Carnivora	92	8.4	9.0	4	7.5	12.1	–0.9	–3.1
Pinnipedia	18	1.6	1.8	0	0	0	–1.6	–1.8
Hyracoidea	3	0.3	0.3	0	0	0	–0.3	–0.3
Perrisodactyla	6	0.5	0.6	0	0	0	–0.5	–0.6
Artiodactyla	81	7.4	7.9	3	5.7	9.1	–1.7	+1.2
Total placentals	1,021	92.9		33	62.3			
Total mammals	1,099			53				

* >5% deviation of sample from actual.

To begin with, primates are over-weighted by 4.8%. This overweighting of primates is also no accident. Since another of the goals of this project is to trace the evolution of the ascending auditory system along the human lineage, and since most of the ancestors in the human lineage are primates, this extra number of primates will be justified in a future report.

For the purpose of judging the representativeness within the placental sample, the most serious deviations are found in rodents and Chiroptera (bats). The sample under-represents rodents by 12.4% and Chiroptera by 15% in terms of total mammals (Col. 7) and under-represents Chiroptera by –15.2% and over-represents primates by 11.2% in terms of total placentals (Col. 8). In terms of numbers of genera, there is one genus of bats in the sample, whereas there should have been 8; there are 14 genera of rodents in the sample, whereas there should have been 18.

Turning first to the under-representation of the Chiroptera, with only one genus representing the 186 diverse genera within the order of Chiroptera, it is notable that the single genus appears as a 'deviant' in some of the present analyses nevertheless. Although it is tempting to conclude that, in general, bats are deviant because of their peculiar use of the auditory system in echo-location, a study focused on bats alone is required before any conclusions about bats relative to other mammals can be taken seriously. Therefore, the present data on bats are best used only as a benchmark, certainly not as a representation of all Chiroptera.

Finally turning to the under-representation of rodents, 426 (or more than 40%) of the 1,021 genera of placentals in

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the 14 orders are rodents. The present sample contains about 75% of the number that should have been included in the sample if it were to be truly 'representative' (i.e., 14 genera vs. 18 genera).

Furthermore, one of the rodents, namely the laboratory mouse, stands out as one of the deviant animals in the sample. That is, for this mouse, the 'usual form'accounts for only one-half of the total variance among the nuclei of its auditory system. There is an obvious reason why this result might be misleading. The single 'mouse' representing the one genus, *Mus*, is a 'laboratory' mouse – that is, an unidentified inbred strain. Therefore, the deviance of this particular mouse may be peculiar to the inbred strain and not to the species or genus. For this reason, it is not safe to conclude that the genus Mus is necessarily deviant, even though the deviation of the present mouse is very large. Tempering this argument, however, it is interesting to note that many of the 13 other genera of rodents in the sample are also 'unusual' in one way or another: the sample includes the water-dwelling beaver, the very large capybara, the poor-hearing pocket gopher [Heffner et al., 1994], the desert-dwelling kangaroo rat [Webster and Webster, 1971], the fossorial mole rat [e.g., Bruns et al., 1989] and the inbred albino rat. Yet these same genera, despite their peculiar heritage or habitus, are more notable for their similarity to the usual form than for their deviation.

Therefore, though only 2 of the 14 orders of placentals in the sample vary from the actual worldwide distribution by more than 5%, they vary widely enough that the present sample of placentals can *not* be generalized without caveat to the population of placental mammals nor even to the subpopulation of non-echolocating placental mammals.

In sum, despite the relatively good representation within the marsupial sub-sample, the present results can not be confidently generalized to the population of viviparous mammals. Nevertheless, because the 'usual forms' calculated here, are relatively immune to great change (due to the numerically large size of the present sample), they probably provide a reasonable and appropriate bench-mark to which still other mammals can be compared as measurements of their auditory systems become available in the future.

Further Analyses of the Data

Given that the non-representativeness of the present sample of mammals can be only cautiously generalized, and then only to non-echolocating mammals, at least three questions arise as to its usefulness in hypothesis construction for functional experimentation. As noted previously, the usefulness of the present data for seeking direct structure-function correspondence probably must await further treatment – first, to remove the possibly lasting effects of selective pressures present in the remote geological past which are not necessarily still present today. As mentioned above, such ancient structural modifications are probably present in the data, insofar as there are statistically reliable inter-subclass differences – that is, between the marsupials and the placentals in the sample – and *not* necessarily between lesser-level taxa within either marsupials or placentals. The further parcellation of historical adaptations from more recent adaptations through taxonomic analysis is the topic of a future report.

Second, in seeking structure-function correspondence in the present data, it should be noted that some of the nuclear-level deviations evidenced in table 4 may have been driven by selective pressures only remotely related to the strictly auditory features of the environment. For example, as mentioned previously a simple one-way analysis of variance of the (log-normalized) data in table 3 shows that MG is a larger percentage of the total auditory system in placentals than it is in marsupials (25 vs. 16%). Though statistically reliable ($p < 0.01$), this difference may be a secondary effect of a more pervasive selective pressure, perhaps on isocortical representation of sounds or sound sources or even by a still more generalized pressure for wider isocortical representation of each of the sensory systems. In this case the expansion of MG might be a secondary result of an expansion of auditory cortex and have only a remote and possibly undetectable relationship with changes in the selective pressures exerted by the auditory environment itself.

Generalizing this problem to the entire data set, it is more likely than not that selective pressures for the use of ear-transduced environmental cues are greater for nuclei synaptically closer to the ear (e.g., cochlear nucleus) than to nuclei synaptically more remote from the ear (e.g., medial geniculate or auditory cortex). If this idea of synaptic insulation from environmental vagaries is correct, it is correspondences between the form of the cochlear nuclei and current environmental demands that might be easier to find because these nuclei might have responded more quickly to changes in strictly auditory demands and hence reflect current ecologies. In this case, statistically reliable differences among the cochlear nuclei should be present between families within one or another order or even between genera within one or another family. This possibility also can be examined with hierarchical analysis of the represented taxa – again, the subject of a future report.

The third problem contained in the present data surrounds the use of a nucleus in its entirety as the unit of analysis. This choice of unit obviates statistical analysis of more refined sub-parts, such as subdivisions of each nu-

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cleus based on like cell-type, or like afferent supply, or like efferent target, etc. This problem is not just one of philosophy. Although 'unit-of-analysis' is interchangeable mathematically, structure-function parallels already discovered in the CNS have shown that each of these alternatives often have a more direct functional correspondence than entire nuclei. Indeed, the usefulness of some of these alternatives stands in contrast to the present data for some of the nuclei. For example, MTB (probably in all mammals) and LSO (probably in all rodents) give rise to many olivocochlear fibers [e.g., Robertson, 1985; Warr, 1992]. Hence, their constituent cells are not all part of the ascending system. Nevertheless, the contributions of the olivocochlear cells to the volume of the two respective nuclei were *not* subtracted from the volumes in table 1 because they can not be reliably distinguished in Nissl-stained material. Similarly, the inferior colliculus and medial geniculate are themselves complexes of a wide variety of cell types, each with their own capacity for integration and differentiation of the afferent neural activity reaching them [e.g., Morest, 1964, 1965, 1971, 1975; Aitkin and Webster, 1972; Aitkin, 1973; Ryugo and Killackey, 1974; Kudo and Niimi, 1978; Winer and Morest, 1983; Morest and Oliver, 1984; Winer, 1984, 1985; Imig and Morel, 1985; Winer and Larue, 1987]. Combining each of these complexes into only one 'nucleus' may well annihilate several specific structure-function correspondences more useful than those discoverable with the present, less refined data.

As another example of this pervasive problem, DCN is known to be an integrative network in its own right with many different cell types and many intrinsic interconnections, despite the simple-appearing extrinsic afferents and efferents of the entire nucleus. Aitkin and Nelson [1989], Moore [1980], Osen and Mugnaini [1980] [also see Voigt and Young, 1980; Schwartz, 1981; Adams, 1983; Manis and Brownell, 1983; Rhode et al., 1983; Blackstad et al., 1984; Ryugo and Willard, 1985; Spirou et al., 1991; Berrebi and Mugnaini, 1991] have each made this point previously, and their separate analyses of DCN may help explain the variation in the size of DCN seen in the present data. For example, Aitkin and Nelson [1989] have shown that the huge DCN of feathertailed gliders seems to contain a large volume of unique cell types not seen in other mammals – sufficiently different to warrant a special name – 'lateral lobe'. Similarly, Moore [1980] has shown what seems to be an almost progressive atrophy in DCN along the primate segment of the human lineage.

These considerations, taken together, yield a host of potentially useful alternatives to the present data set based on entire nuclei alone. Although each would seem to entail an heroic effort in data collection, the Nissl-stained brain sections used in this study were not destroyed or degraded in the process of the present data collection. Hence, they still remain available to any wishing to use them to extract other data based on a more refined unit of analysis.

Therefore, the final goal of the present project – to supply structure-function correspondences that can suggest at least global functions for auditory nuclei – can not yet be fully realized. At present we plan further analyses in companion reports to bring this final goal nearer to fruition. Nevertheless, the present data set and the conclusions based on it should be useable for other, more practical, purposes.

Acknowledgements

Supported in part by NIH grants NSO7726 and DC00197. The authors thank: Drs. L. Aitkin and J. Nelson, Monash University, for making their marsupial materials available to us; Dr. Este Armstrong at the Armed Forces Institute of Pathology in Washington for giving us access to the Yakovlev Collection for the human material; Dr. Wally Welker, University of Wisconsin, Department of Neurophysiology, for allowing us access to their placental materials; Drs. R.S. and H.E. Heffner for rodent materials; Dr. Donald Wong, Indiana University, for bat body and brain weights; Dr. J. Brunso-Bechtold and Dr. R. Bechtold, Wake Forest University, Bowman Gray School of Medicine, for systematic photography of the University of Wisconsin materials; Ms. J. Persons for morphometric measurements of many of the placental materials, and Dr. Lindsay Aitkin for his comments on an earlier version of this manuscript.

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Erratum

In reference to 'Responses to Moving Small-Field Stimuli by the Praying Mantis, *Sphodromantis lineola* (Burmeister)', by F.R. Prete and T. McLean, published in Brain Behav Evol 1996;47:42–54, the journal has been asked to print the following erratum. The name of Patricia J. Placek McMillin should be added as an author of this paper, and the title page should indicate that a portion of the work was done at Youngstown State University, Youngstown, Ohio, USA.

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